Report

SPDC-BFPC UPSTREAM GAS SUPPLY, GATHERING, EVACUATION & PROCESSING (UGS-GEP) PROJECT

Pipeline Transient Simulation Study

Gas Gathering & Evacuation Pipelines

Prepared for:

Brass Fertilizer & Petrochemical Company Limited

Doc. No. EA/SPDC-BFPC UGS-GEP/PRC-RPT-001

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SUMMARY PAGE

Client:	BRASS FERTILIZER & PETROCHEMICAL COMPANY LIMITED	
Project:	SPDC-BFPC UGS-GEP	**Title**: Pipeline Transient Simulation Study
Facilities:	Gas Gathering & Evacuation Pipelines	**Project Ref: EA/SPDC-BFPC UGS-GEP/PRC-RPT-003**

The administration of this document is in respect of the Gas Gathering & Evacuation Pipelines aspects of the SPDC-BFPC Upstream Gas Supply, Gathering, Evacuation and Processing (UGS-GEP) Project.

The document defines the basic design elements as basis for the execution of the Front End and Engineering Design (FEED) that will lead to the project's FID and implementation, as well as demonstrates readiness to progress to the FEED as the next phase.

It is also a presentation to DPR as a statutory requirement in the process of securing regulators' approval to progress the FEED / Detail Design that will lead to the project's FID and implementation.

The report has relied and used data and reference documents provided by client, and if further document and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the concept selection reports that constitute the basis for the basic design.

KEY WORDS: BRASS, DESIGN, ENGINEERING, EXECUTION, FERTILIZER, GAS, PLANT, READINESS.

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Abbreviations

The Following definitions and abbreviations have been used in this document.

Organisational

BFCL	Brass Fertilizer Company Limited
EA	Epic Atlantic Limited
SPDC	The Shell Petroleum Development Company of Nigeria Limited

Technical Terms

BBOU	Bubuowei Bou
NUNR	Nun River
SEIB	Seibou
ELEP	ELEP
OPUG	Opughene
AKAB	Akabel
T/L	Truckline
B/L	Bulkline
UGS	Upstream Gas Supply
GEP	Gathering, Evacuation, Processing
FEED	Front End Engineering Design
FID	Final Investment decision
NAG	Non Associated Gas
CTP	Custody Transfer point
MF	Manifold
GCR	Gas Condensate Ration

Units of Measurements

mm	milimeter
m	meter
km	kilometer
m^2	Square meter
m^3	Cubic meter
MMscfd	Million Standard Cubic Feet per Day
m/s	Meter per second
kg/ m^3	Kilogram per cubic meter
bara	Pressure
°C	Degree celciuus
MTPA	Metric Tonnes Per Annum
DWT	Deadweight tonnage

1 Administration

This Conceptual Study Memorandum is in respect of the SPDC-BFPC Upstream Gas Supply, Gathering, Evacuation and Processing (UGS-GEP) Project. It stems from the Concept Select Studies, and presents the selected design concepts as basis for the Basic Design of the project.

It is also (in conjunction with the Basic Design) a presentation to DPR as a statutory requirement in the process of securing regulators' approval to execute the FEED / Detail Design that will lead to the project's FID and implementation.

The report has relied and used data and reference documents provided by client, and which to a large extent is adequate for this phase of work: - conceptual study and basic design. However, if further information and / or refinement become available, Epic Atlantic reserves the right to update accordingly, the opinion set out in this report.

Furthermore the security situation in the region with its wide spate of vandalism may dictate that certain design provisions be changed during the FEED. The situation will continue to be monitored, and if still perceived to be a cog in the wheel of innovation, some of the ideal designs will be replaced with what will be practically suitable for the situation. For example, in as much as the concept for power cables to the manifold platforms to provide power for automation is ideal, there will be no need to do that if the facilities will eventually be vandalised in no time. Instead, it will be rather more reasonable to change the design to manual valve operations, and without the features of automation that will require electricity.

1.1. Background and Introduction

The SPDC-BFPC Upstream Gas Supply, Gathering, Evacuation and Processing (UGS-GEP) Project is constituted by Brass Fertilizer & Petrochemical Company Limited (BFPC), in partnership with The Shell Petroleum Development Company of Nigeria (SPDC).

The project scope includes Gas Gathering Pipelines, and a Gas Process Processing Plant that will provide processed gas as feedstock to a Petrochemical Plant and a Power Plant. The Gas Processing Plant will process natural gas gathered from five SPDC fields – Bubouwe Bou, Elepa, Nun River, Seibou, and Opugbene - in OMLs 33, 32, & 36.

The Gas Plant will comprise three trains, (each with a processing capacity of 150 MMscfd) and condensate storage capacity of 50,000 Tonnes. In this respect, BFPC have concluded the Gas Supply & Processing Agreement (GSPA) with SPDC for 270 MMscfd of the natural gas feedstock. The Plants (Gas Plant, Petrochemical Plant, and

Power Plant) will all be located in the same complex at Akabel, Brass Island, Bayelsa state of Nigeria, near the shores of Atlantic Ocean.

The Petrochemical Plant will comprise a Methanol Plant of 1.75 MTPA capacity, and a Urea Plant of 2.6 MTPA total capacity, built in two phases of 1.3 MTPA each. It will also have storage capacity for 150,000 Metric Tonnes of Methanol and 100,000 Metric Tonnes of Urea.

The facilities complex will also comprise a captive export jetty of 4 berths for up to 35,000 DWT vessels.

The current on-stream date for the project is June 2020, and to achieve that, FID is planned for December, 2017. In pursuit of these business objectives, BFPC have virtually concluded Project Management & EPC contracts with EIL, Saipem, and TCC, who are internationally renowned contractors in the implementation of the subject projects type.

The Gas Processing Plant, and the associated Gas Gathering System / Pipelines which will transport wellhead gas from the SPDC fields to the Gas Processing Plant, constitute the key upstream enabler facilities for the realisation of the objective supply of processed gas supply as feedstock to the Petrochemical Plant for the product manufacturing of methanol & urea, and to the Power Plant for power generation.

1.1 Scope of Document

The scope of the document covers the Pipeline Transient Simulation Study necessary to provided adequate information and data to establish the basic design and to validate & progress the project to the next phase of FEED / Detail Design:

2 Objectives

The objectives of this document are to:

- Establish minimum turndown rate and verify the effect of different production rates on the inlet pressure, inlet temperature, liquid hold-up, slug and liquid surge for the pipelines.
- Check hydrates formation potential of the pipelines after shutdown.
- Perform ramp-up analysis from turndown rates and determine surge volume.
- Perform pipeline pigging operation to determine surge volume and establish the allowable pigging flowrate considering manufacturers' pig tool maximum velocity.
- Perform shutdown and start-up simulations and determine surge volume due to pipeline start-up.
- Perform slugging analysis.

2.1 Battery Limit

For simulation, the battery limit of individual NAG Pipeline (BBOU, ELEP) starts from downstream tie-in of the Metering Unit at the Custody Transfer Point to the pipeline tie-in upstream of the NAG inlet manifold at the AKAB Gas Processing Plant.

2.2 NAG Pipeline Schematic Representation

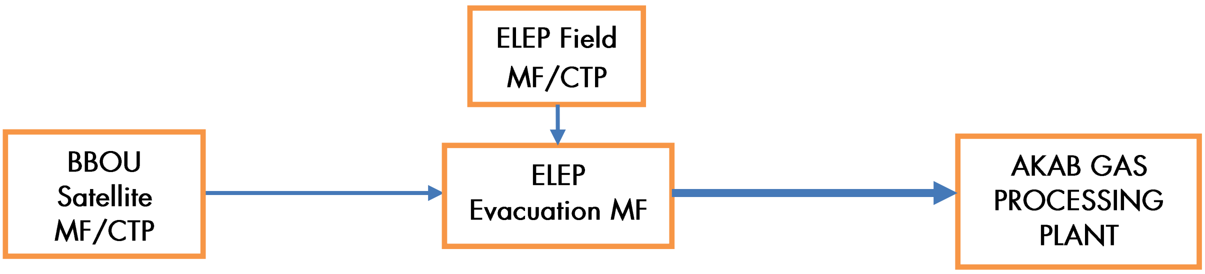

Fig 2.1. Pipeline Simulation Scope

2.3 Unit

The units shown below (SI units in general) shall be used as the main design units and all values quoted shall be in these units. It is permissible to show alternative values and units as indicated in the brackets, after the design unit, when it is considered desirable. The size of the unit shall be appropriate to the value being measured, e.g. mm - m - km.

Table 2.1: Unit Description

Description	Unit
Nominal pipe diameters	mm (inch)
wall Thickness	mm (inch)
Length	m (mm or km)
Area	m^2
Liquid Volume	m^3
Volumetric Flow Rate	m^3/d
Standard Gas Volumetric Flow Rate	MMscfd
Velocity	m/s
Density	kg/m^3
Pressure	bar
Temperature	°C
Viscosity	cP

3 Methodology

3.1 Introduction

OLGA version 7.0 transient simulator was used for building and analysing models for the Gas Gathering Pipelines. The fluid composition was made suitable for use by OLGA via the use of PVTsim to create tab files, which transfer fluid thermodynamic properties into OLGA software in the form required for its use.

The model assumes that inlet to the system is a closed node and a mass source that provides the specified flowrates is placed close to the inlet node.

Since the processing facility conditions are known and specified at the outlet node of the model the simulator is able to compute the required parameters along the line length down to the inlet nodes, such that the entire process from the inlet to the processing facility is fully defined.

3.2 Fluid Composition

NAG reservoir fluids were provided as dry gas compositions as shown in table 3.1 below. The hypo components are shown in table 3.2

Table 3.1: NAG Reservoirs Composition

Component	Compositions (Mole Percent)			
	BBOU Reservoir	ELEP Reservoir	NUNR Reservoir	SIEB Reservoir
Nitrogen	0.07	0.00	0.00	0.07
CO_2	0.59	2.22	0.77	1.55
H_2S	0.00	0.00	0.00	0.00
C1	92.92	81.66	72.83	78.62
C2	3.16	6.13	7.68	5.93
C3	1.01	2.59	4.85	3.40
i-C4	0.61	0.75	1.46	1.22
n-C4	0.44	0.81	1.97	1.38
i-C5	0.25	0.45	0.95	0.83
n-C5	0.19	0.33	0.76	0.58
C6	0.23	0.65	1.03	0.92
C7	0.19	0.90	1.41	1.13
C8	0.04	0.87	1.61	1.14
C9	0.03	0.38	0.90	0.65
C10	0.02	0.30	0.65	0.51
C11	0.03	0.23	0.47	0.30
C12+	0.22	1.73	2.66	1.82
Total	100.00	100.00	100.00	100.00

Table 3.2: Hypo Components

Hypo Component	Density (kg/m^3)				Molecular Mass ($gmol^{-1}$)			
	BBOU	ELEP	NUNR	SIEB	BBOU	ELEP	NUNR	SIEB
C7+	811	828	813	817	167.69	186.72	176.38	174.90

However, this composition is saturated at the reservoir conditions and adjusted to field GCR to obtain the composition shown in table 3.3 which was used as input data for the simulation work.

Table 3.3: NAG Reservoir Compositions - Saturated Stream Condition

Component	Compositions (Mole Fraction)			
	BBOU Reservoir	ELEP Reservoir	NUNR Reservoir	SIEB Reservoir
H_2O	0.0034	0.0105	0.0059	0.0092
Nitrogen	0.0007	0.0000	0.0000	0.0070
CO_2	0.0057	0.0210	0.0076	0.0148
H_2S	0.0000	0.0000	0.0000	0.0000
C1	0.9052	0.7729	0.7167	0.7496
C2	0.0308	0.0581	0.0756	0.0566
C3	0.0099	0.0247	0.0478	0.0326
i-C4	0.0060	0.0072	0.0144	0.0118
n-C4	0.0043	0.0078	0.0195	0.0134
i-C5	0.0025	0.0045	0.0095	0.0083
n-C5	0.0019	0.0034	0.0076	0.0059
C6	0.0002	0.0074	0.0106	0.0105
C7+	0.0293	0.0824	0.0848	0.0866
Total	1.0000	1.0000	1.0000	1.0000

3.3 Design Flowrates and Inlet Temperature

Table 3.4: Pipeline Design Summary

Line Description	Arrival Pressure (bara)	Flowrate (MMscfd)	Nominal Pipe Size (in)	Wall Thickness		Approx. Length (km)
				(mm)	(in)	
BBOU to ELEP	-	180/300	16	22.225	0.875	9.0
ELEP to AKAB	65	300	24	31.75	1.25	28.5

3.4 Pipeline and Coating Materials

The material properties used in the simulations are listed in Table 3.5.

Table 3.5: Pipeline and Coating Materials Properties

Material	Density (kg/m3)	Heat Capacity (J/kg-K)	Thermal Conductivity (w/m-K)	Thickness (mm)	Roughness (mm)
Carbon Steel	7,800	500	45	22.225 / 31.75	0.03
3LPE	900	2,300	0.4	3	-
Wet Sand	2,500	880	2.1	1,000	-

3.5 Parameter Descriptions

The key parameters for slug tendency analysis are as follows:

Table 3.6: Parameters Description

Parameter	Description
PT	Pressure
TM	Temperature
ID	Flow regime indicator
HOL	Holdup (Liquid volume fraction)
QLT	Total liquid volume flow
ACCLIQ	Accumulated liquid volume flow
EVR	Erosion velocity ratio
GLRST	Gas liquid ratio at standard condition
LIQCFR	Total liquid volume fraction
LIQC	Total liquid volume in branch
DTHYD	Difference between hydrate and section temperature
SURGELIQ	Surge liquid volume
NSLUG	Total number of slugs in the pipeline

Table 3.7: OLGA Numbers For Flow Regime Indication

Number	Flow Regime

1	Stratified
2	Annular
3	Slug
4	Bubble

The OLGA number provides information on flow pattern - slug tendency, liquid surge flow rate, and liquid content along the pipeline, HOL, QLT, ACCLIQ, and changes of LIQCFR with time, etc - which are parameters of influence in the design of slug catcher.

3.6 Pipeline Elevation

Pipeline elevation used is assumed to be level. Real pipeline profile will be used after route survey has established pipeline profile and elevation.

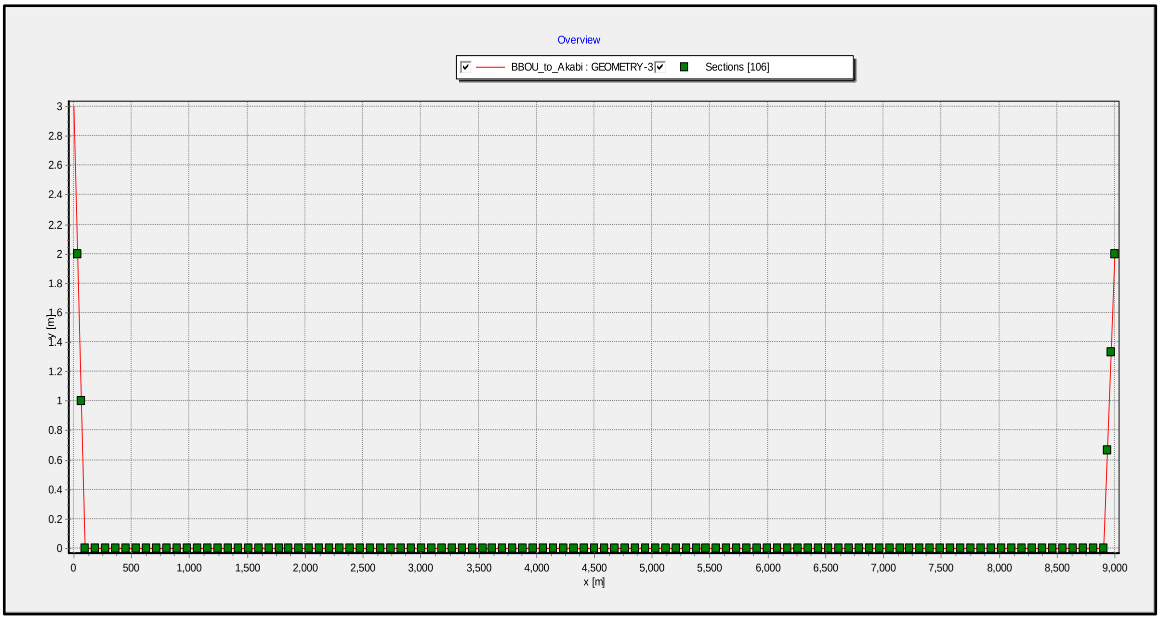

Fig 3.1: BBOU to ELEP Pipeline Profile.

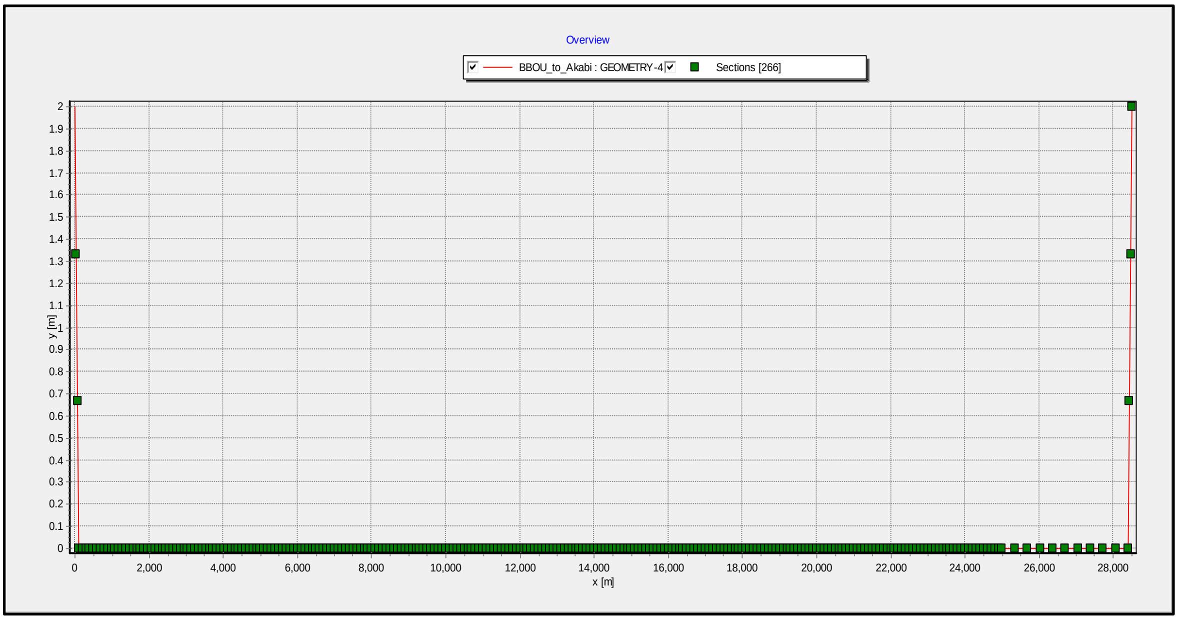

Fig 3.2: ELEP to AKAB Pipeline Profile

4 Steady State Simulation Results

4.1 Preliminary Line Sizing

4.1.1 BBOU to ELEP at 300 MMscfd

The preliminary line sizing simulation was carried out based on peak flowrate of 300 MMscfd for BBOU to Elepa pipeline at different line sizes of 10", 12", 14", 16", 18" and 20" to determine the minimum line size for BBOU to ELEP pipeline.

Table 4.1: Preliminary pipeline result for BBOU to ELEP

	BBOU to ELEP Pipeline			
Case	Flowrate (MMscfd)	Pipe Size (in)	Inlet Pressure (bars)	Gas Outlet Velocity (m/s)
1	300	10	120.00	28.06
2	300	12	120.00	18.86
3	300	14	115.23	15.30
4	300	16	99.52	11.63
5	300	18	91.70	9.14
6	300	20	87.92	7.38

From the result on table 4.1, the minimum line size for BBOU to ELEP pipeline is 16".

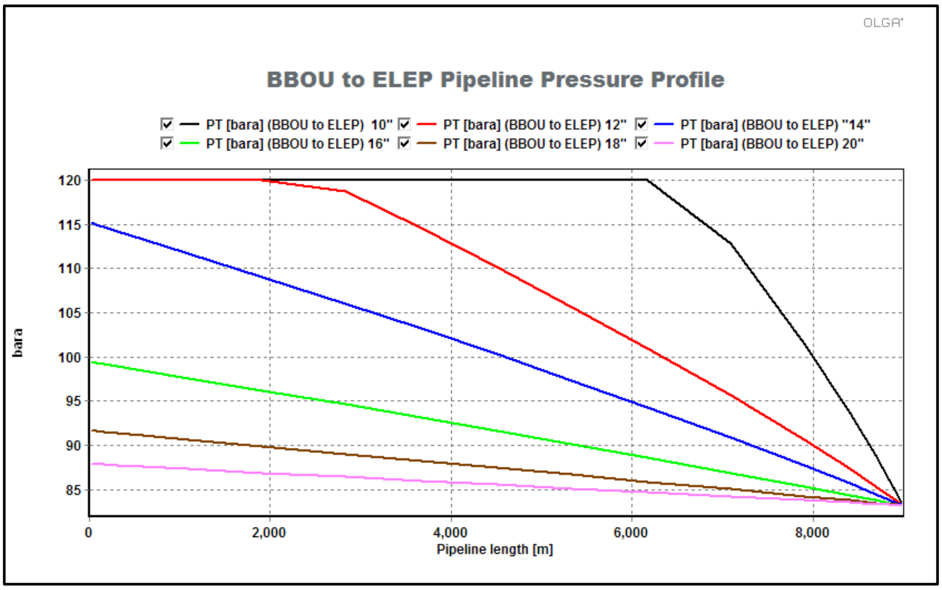

Fig 4.1: BBOU to ELEP inlet pressure profile plot at different line sizes.

Table 4.2: Preliminary pipeline result

	BBOU to ELEP Pipeline				ELEP to AKAB Pipeline			
Case	Flowrate (MMscfd) (BBOU)	Pipe Size (in)	Inlet Pressure (bars)	Gas Outlet Velocity (m/s)	Flowrate (MMscfd) (BBOU + ELEP)	Pipe Size (in)	Inlet Pressure (bars)	Gas Outlet Velocity (m/s)
1	180	16	98.97	6.78	300	20	93.25	11.06
2	180	16	88.77	7.70	120	22	82.29	9.22
3	180	16	82.94	8.34	120	24	75.90	7.75
4	180	16	79.53	8.77	120	26	75.12	6.41
5	180	16	77.39	9.07	120	28	69.74	6.02

Based on maximum inlet pressure of 100 bara at BBOU metering point and 80 bara at ELEP metering point, the minimum pipe sizes are 16" for BBOU to ELEP pipeline and 22" for ELEP to AKAB pipeline. However 24" is suitable considering the capacity up to 400 MMscfd target

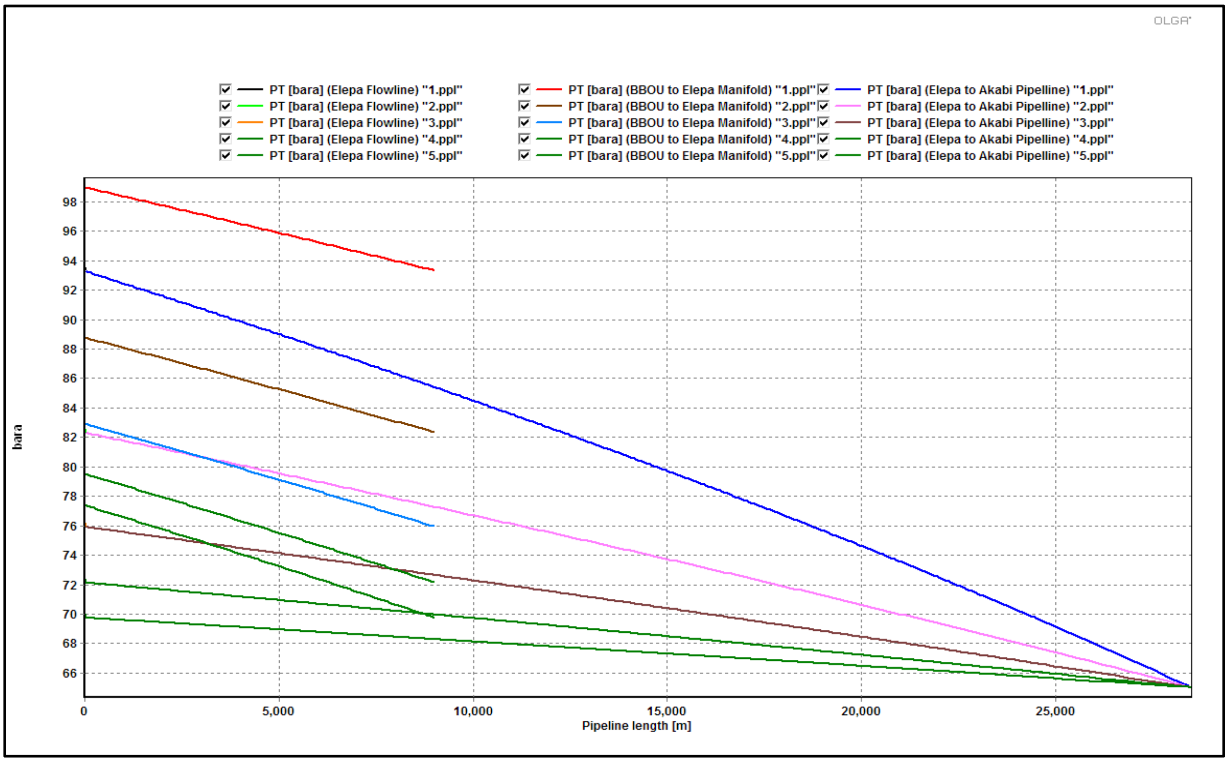

Fig 4.2: BBOU to ELEP and ELEP to AKAB Inlet Pressure Profile Plot at different line sizes.

4.2 Minimum Turndown Simulation Result

Three studies were carried out to determine the minimum turndown flowrates for 16"x 9 km BBOU to ELEP pipeline and 24"x 28.5 km ELEP to AKAB pipeline. Case 4 is not included in minimum turndown calculation since it has been covered in case 1.

- Case 1: 180 MMscfd flowing from BBOU field with no contribution from ELEP field. Minimum turndown was determined for the 16' BBOU to ELEP and 24" ELEP to AKAB pipeline
- Case 2: 120 MMscfd flowing from ELEP field with no contribution from BBOU. Minimum turndown was determined for only 24" ELEP to AKAB pipeline.
- Case 3: Combined flow from BBOU and ELEP fields. BBOU contributes 180 MMscfd and ELEP contributes 120 MMscfd
- Case 4: 300 MMscfd flowing from BBOU field with no contribution from ELEP field. Minimum turndown was determined for the 16' BBOU to ELEP and 24" ELEP to AKAB pipeline.

4.2.1 Case 1: : 180 MMscfd flowing from BBOU field

Minimum Turndown for 16" BBOU to ELEP Pipeline for case 1.

Table 4.3: Flowrate/ Inlet Pressure

Flowrate (MMscfd)	Inlet Pressure (bara)
1	65.3265
2	65.3215
3	65.3154
4	65.3096
5	65.3044
6	65.3001
7	65.2969
8	65.2947
9	65.2935
10	65.2933
11	65.2942
12	65.2959
13	65.2986
14	65.3022
15	65.3067
16	65.3119
17	65.3179
18	65.3248
19	65.3323
20	65.3402

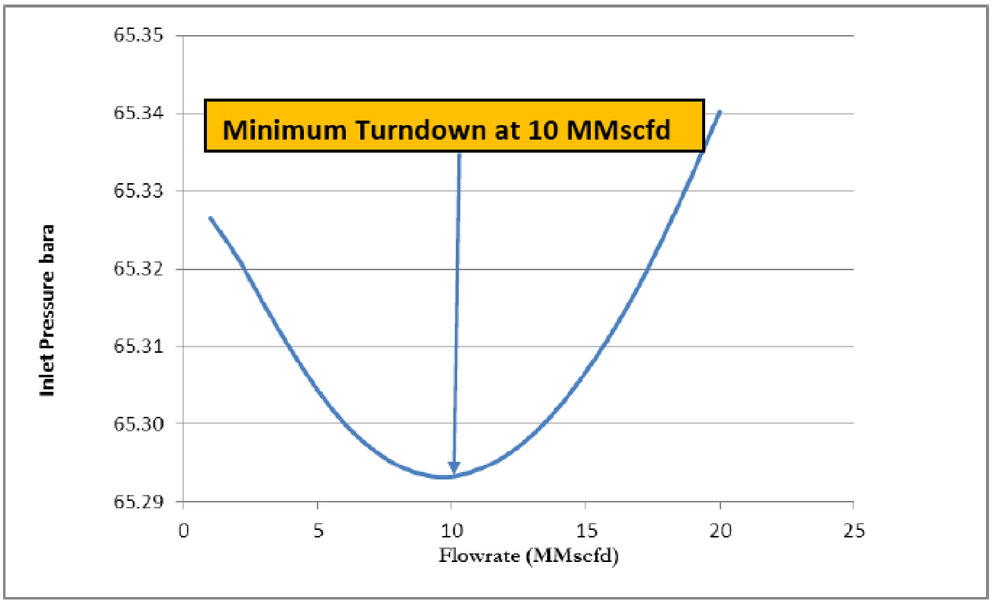

Fig 4.3: Steady State Minimum Turndown Estimation for 16" BBOU to Elepa

Table 4.4: Flowrate/ Inlet Pressure

Flowrate (MMscfd)	Inlet Pressure (bara)
1	65.1597
2	65.1608
3	65.1612
4	65.1610
5	65.1607
6	65.1602
7	65.1597
8	65.1593
9	65.1590
10	65.1589
11	65.1590
12	65.1593
13	65.1598
14	65.1606
15	65.1615
16	65.1626
17	65.1638
18	65.1653
19	65.1670
20	65.1684

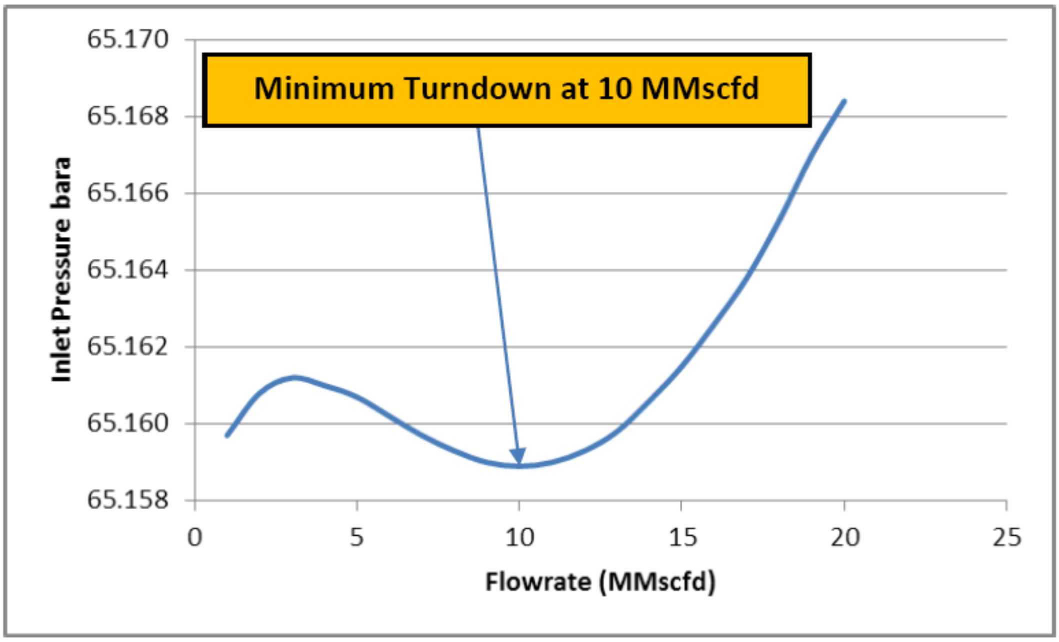

Fig 4.4: Steady State Minimum Turndown Estimation for ELEP to AKAB NAG Pipeline

4.2.2 Case 2: 120 MMscfd flowing from ELEP field.

Minimum Turndown for 24" ELEP to AKAB Pipeline for case 2

Table 4.5: Flowrate/ Inlet Pressure

Flowrate (MMscfd)	Inlet Pressure (bara)
1	65.1587
2	65.1587
3	65.1584
4	65.1578
5	65.1571
6	65.1566
7	65.1564
8	65.1566
9	65.1572
10	65.1583
11	65.1597
12	65.1614
13	65.1635
14	65.1659
15	65.1693
16	65.1718
17	65.1747
18	65.1779
19	65.1815
20	65.1854

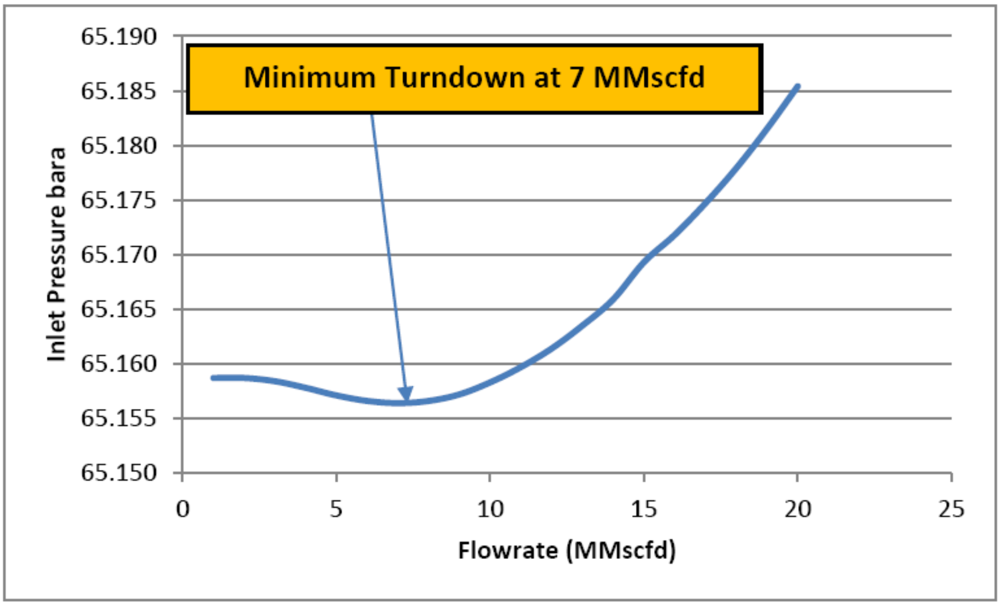

Fig 4.5: Steady State Minimum Turndown Estimation for Elepa to Akabi NAG Pipeline

4.3 General Steady State Result

General Steady State result was reported for the four cases stated in 4.2.

Table 4.6: Case 1_ BBOU to ELEP 16" pipeline steady state results.

Flowrate (MMscfd)	Inlet Pressure (bara)	Outlet Temperature (°C)	Gas Velocity (m/s)	Erosional Velocity Ratio (-)	Pipeline Water Content (m^3)	Pipeline Condensate Content (m^3)	Total Liquid Holdup (m^3)	Water Volume Flow (m^3/d)	Condensate Volume Flow (m^3/d)	Total Liquid Volume Flow (m^3/d)
5	65.3044	58.2710	1.277	0.016	138.402	81.414	219.816	0.390	35.498	35.888
10	65.2933	58.2722	1.554	0.033	41.355	73.364	114.718	0.780	70.995	71.775
20	65.3402	58.2610	2.107	0.065	12.476	71.006	83.483	1.560	141.996	143.556
30	65.4614	58.2346	2.659	0.098	6.733	67.512	74.245	2.340	213.018	215.358
40	65.6681	58.1911	3.210	0.131	4.541	59.586	64.127	3.120	284.080	287.200
50	65.9621	58.1292	3.738	0.163	3.620	54.646	58.267	3.900	355.199	359.099
60	66.3315	58.0514	3.948	0.196	3.086	52.485	55.570	4.680	423.363	428.043
70	66.7432	57.9669	4.298	0.228	2.507	52.812	55.320	5.741	497.646	503.387
80	67.2287	57.8694	4.677	0.261	1.973	52.902	54.874	7.095	568.998	576.093
90	67.7832	57.7592	5.083	0.293	1.444	52.896	54.340	7.832	640.459	648.291
100	68.4093	57.6362	5.557	0.348	1.321	52.658	53.979	9.300	712.042	721.342
110	69.1028	57.5015	6.046	0.357	1.200	52.409	53.609	10.686	783.760	794.446
120	69.8630	57.3590	6.538	0.388	1.092	52.135	53.227	11.978	855.614	867.592
130	70.6875	57.2043	7.027	0.420	0.998	51.804	52.802	13.093	927.625	940.718
140	71.0000	57.0403	7.511	0.451	0.911	51.146	52.057	14.015	999.812	1013.827
150	72.5424	56.8649	7.987	0.482	0.834	50.278	51.113	14.797	1072.190	1086.987
160	73.5695	56.6827	8.444	0.513	0.772	49.398	50.170	15.404	1144.770	1160.174
170	74.6344	56.4920	8.887	0.543	0.721	48.500	49.221	15.912	1217.510	1233.422
180	75.7333	56.2932	9.325	0.572	0.680	47.591	48.271	16.391	1290.420	1306.811

Table 4.7: Case 1_ ELEP to AKAB 24" pipeline steady state results.

Flowrate (MMscfd)	Inlet Pressure (bara)	Outlet Temperature (°C)	Gas Velocity (m/s)	Erosional Velocity Ratio (-)	Pipeline Water Content (m^3)	Pipeline Condensate Content (m^3)	Total Liquid Holdup (m^3)	Water Volume Flow (m^3/d)	Condensate Volume Flow (m^3/d)	Total Liquid Volume Flow (m^3/d)
5	65.1614	57.7368	0.454	0.007	2780.73	481.319	3262.049	0.390	38.528	38.918
10	65.1591	57.7400	0.884	0.014	1060.73	638.807	1699.537	0.780	77.053	77.833
20	65.1675	57.7271	1.661	0.027	330.101	592.911	923.012	1.560	154.087	155.647
30	65.1897	57.6928	1.889	0.041	167.703	578.052	745.755	2.339	231.183	233.522
40	65.2300	57.6364	2.111	0.055	102.585	572.015	674.600	3.119	308.285	311.404
50	65.2871	57.5558	2.333	0.069	71.4363	570.300	641.736	3.899	385.418	389.317
60	65.3596	57.4549	2.555	0.082	52.1738	567.297	619.471	4.678	462.598	467.276
70	65.4471	57.3420	2.776	0.096	39.8698	564.993	604.863	5.458	539.814	545.272
80	65.5580	57.2087	2.997	0.110	30.8759	536.229	567.105	6.237	617.082	623.319
90	65.6893	57.0560	3.438	0.123	24.0746	509.477	533.552	7.016	694.406	701.422
100	65.8410	56.8838	3.658	0.137	20.4601	486.914	507.374	7.795	771.793	779.588
110	66.0143	56.6931	3.870	0.151	19.077	467.897	486.974	8.574	849.245	857.819
120	66.2101	56.4872	4.038	0.164	18.1626	452.220	470.383	9.352	926.761	936.113
130	66.4289	56.2616	4.210	0.178	17.3068	439.280	456.587	10.130	1004.340	1014.470
140	66.6738	56.0183	4.322	0.191	16.6117	428.599	445.211	10.908	1081.990	1092.898
150	66.9451	55.7548	4.399	0.205	15.935	419.656	435.591	11.686	1149.150	1160.836
160	67.2446	55.4753	4.545	0.218	15.2754	411.988	427.263	12.463	1224.880	1237.343
170	67.5469	55.1864	4.689	0.232	14.8078	409.795	424.603	13.240	1299.970	1313.210
180	67.8457	54.8900		0.246	14.4222	411.740	426.162	14.691	1393.340	1408.031

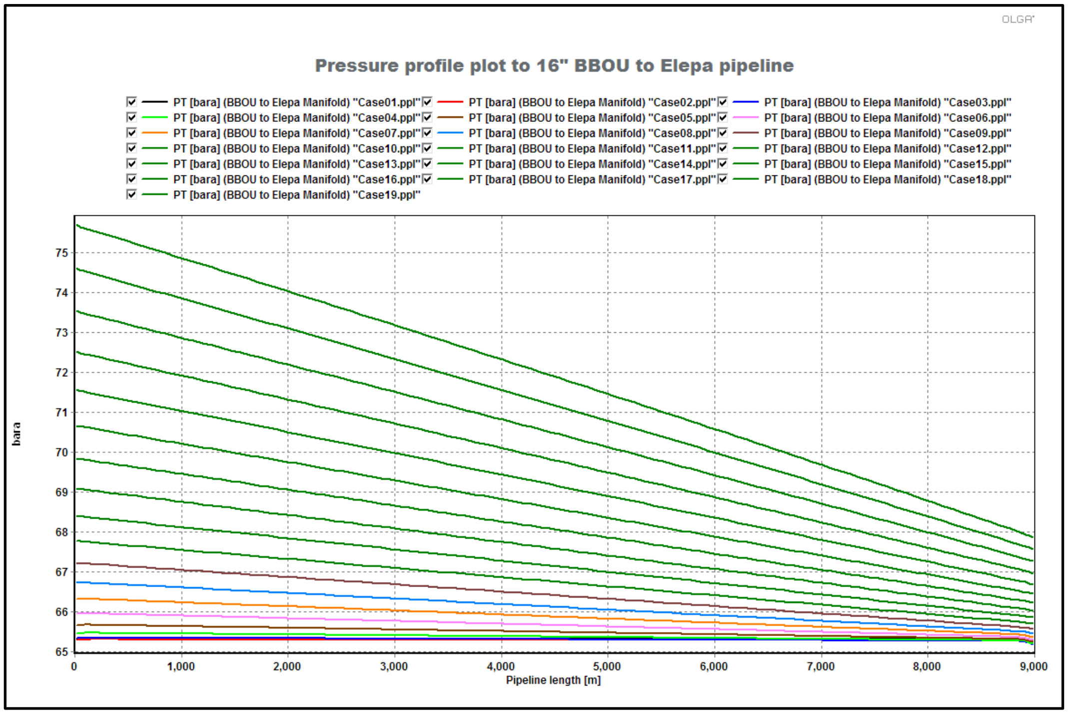

Fig 4.6: Pressure Profile Sensitivity for 16" BBOU to ELEP NAG Pipeline

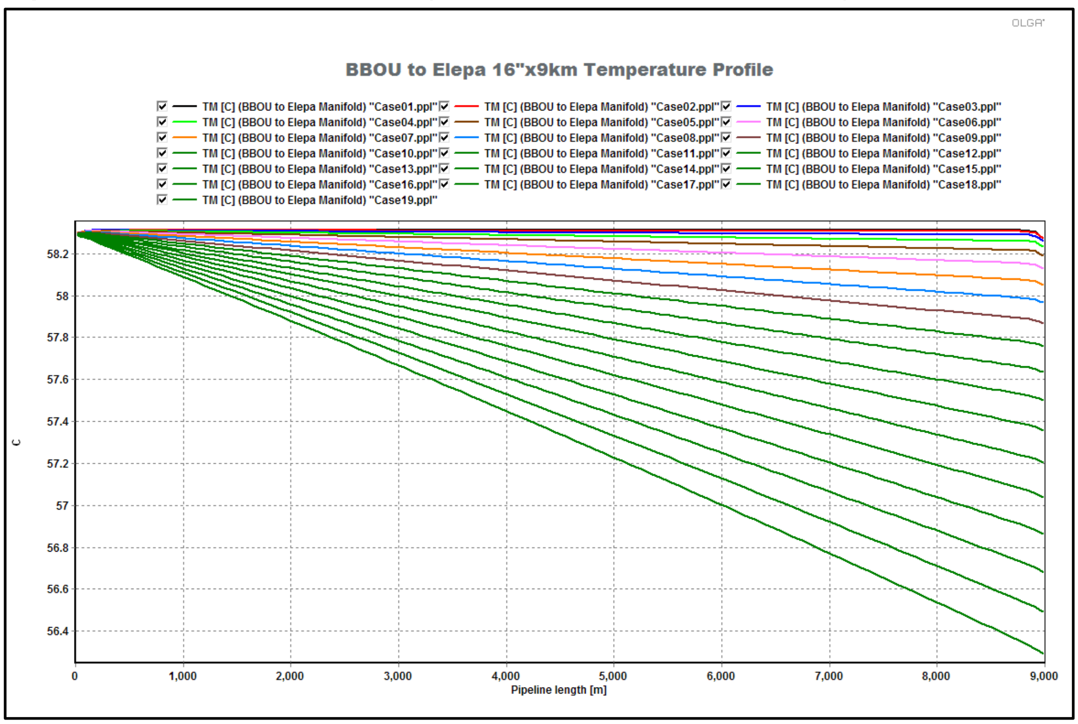

Fig 4.7: Temperature Profile Sensitivity for 16" BBOU to ELEP NAG Pipeline

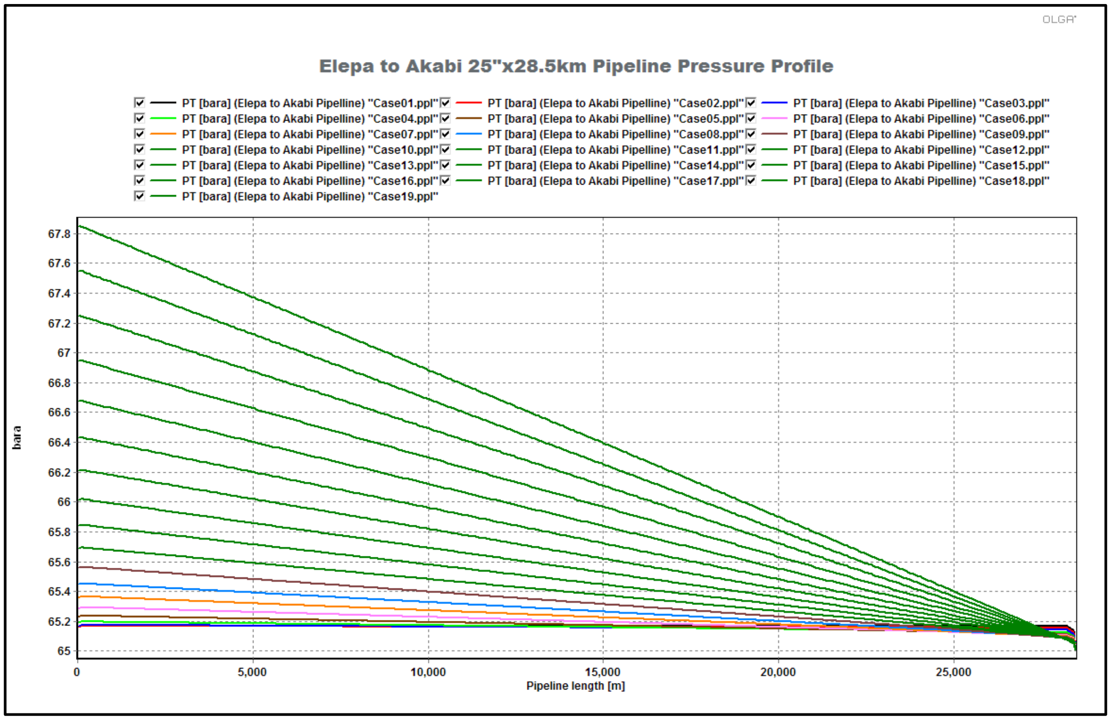

Fig 4.8: Pressure Profile Sensitivity for 24" ELEP to AKAB NAG Pipeline

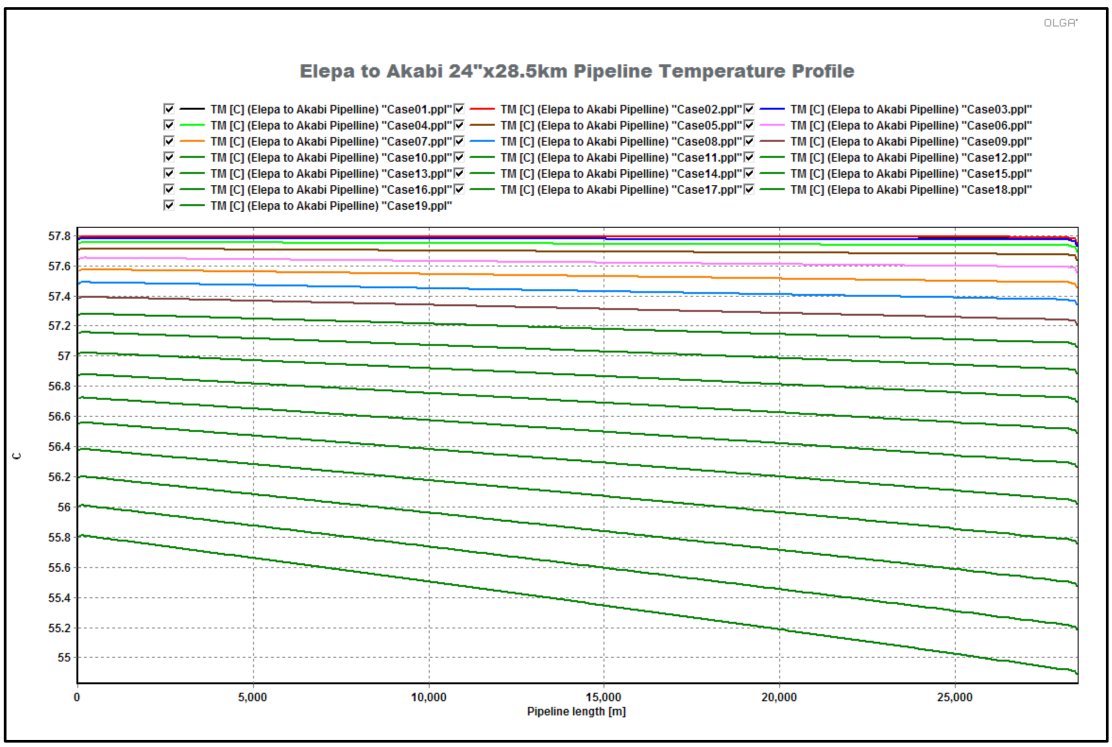

Fig 4.9: Temperature Profile Sensitivity for 24" ELEP to AKAB NAG Pipeline

Table 4.8: Case 2_ ELEP to AKAB 24" pipeline steady state results.

Flowrate (MMscfd)	Inlet Pressure (bara)	Outlet Temperature (°C)	Gas Velocity (m/s)	Erosional Velocity Ratio (-)	Pipeline Water Content (m^3)	Pipeline Condensate Content (m^3)	Total Liquid Holdup (m^3)	Water Volume Flow (m^3/d)	Condensate Volume Flow (m^3/d)	Total Liquid Volume Flow (m^3/d)
5	65.1571	99.9753	0.585	0.010	1419.090	1532.510	2951.600	1.315	148.150	149.465
10	65.1578	99.9776	1.160	0.020	459.383	1427.920	1887.303	2.630	296.285	298.915
15	65.1685	99.9770	1.652	0.030	251.519	1379.210	1630.729	3.945	444.451	448.396
20	65.1840	99.9753	1.794	0.040	163.362	1361.500	1524.862	5.260	592.601	597.861
25	65.2079	99.9721	1.936	0.050	117.777	1350.670	1468.447	6.575	740.754	747.329
30	65.2400	99.9677	2.078	0.060	90.771	1343.120	1433.891	7.890	888.909	896.799
35	65.2797	99.9622	2.220	0.070	73.251	1337.520	1410.771	9.205	1037.068	1046.272
40	65.3268	99.9554	2.362	0.080	61.236	1333.020	1394.256	10.520	1185.235	1195.755
45	65.3812	99.9475	2.504	0.090	52.279	1329.330	1381.609	11.834	1333.403	1345.237
50	65.4428	99.9384	2.646	0.101	45.739	1325.780	1371.519	13.149	1481.587	1494.736
55	65.5124	99.9283	2.788	0.111	40.347	1315.370	1355.717	14.464	1629.772	1644.236
60	65.5941	99.9164	2.929	0.121	36.257	1281.510	1317.768	15.779	1777.965	1793.744
65	65.6848	99.9032	3.071	0.131	34.080	1250.160	1284.240	17.094	1926.176	1943.269
70	65.7849	99.8883	3.213	0.141	32.643	1221.640	1254.283	18.408	2074.395	2092.803
75	65.8948	99.8721	3.355	0.151	31.465	1195.910	1227.375	19.723	2222.623	2242.346
80	66.0150	99.8542	3.497	0.161	30.413	1172.760	1203.173	21.038	2370.868	2391.906
85	66.1458	99.8347	3.638	0.171	29.440	1151.920	1181.360	22.352	2519.130	2541.482
90	66.2878	99.8134	3.780	0.181	28.508	1133.230	1161.738	23.667	2667.410	2691.077
95	66.4412	99.7902	3.906	0.191	27.557	1116.470	1144.027	24.981	2815.716	2840.697
100	66.6064	99.7656	4.015	0.201	26.581	1101.420	1128.001	26.295	2964.039	2990.334
105	66.7846	99.7390	4.125	0.211	25.617	1087.960	1113.577	27.610	3112.379	3139.988
110	66.9764	99.7103	4.237	0.221	24.654	1075.870	1100.524	28.924	3260.745	3289.668
115	67.1821	99.6794	4.350	0.231	23.684	1064.940	1088.624	30.238	3409.137	3439.374
120	67.4019	99.6465	4.463	0.241	22.705	1054.960	1077.664	31.552	3557.563	3589.115

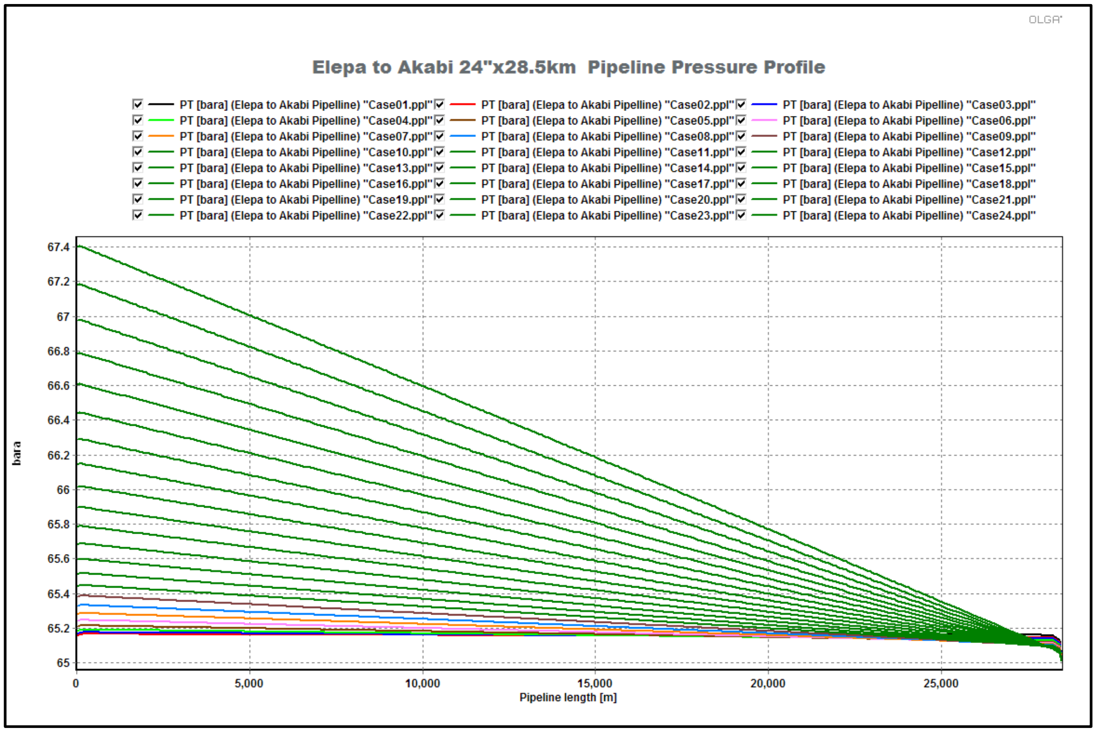

Fig 4.10: Pressure Profile Sensitivity for 24" ELEP to AKAB NAG Pipeline

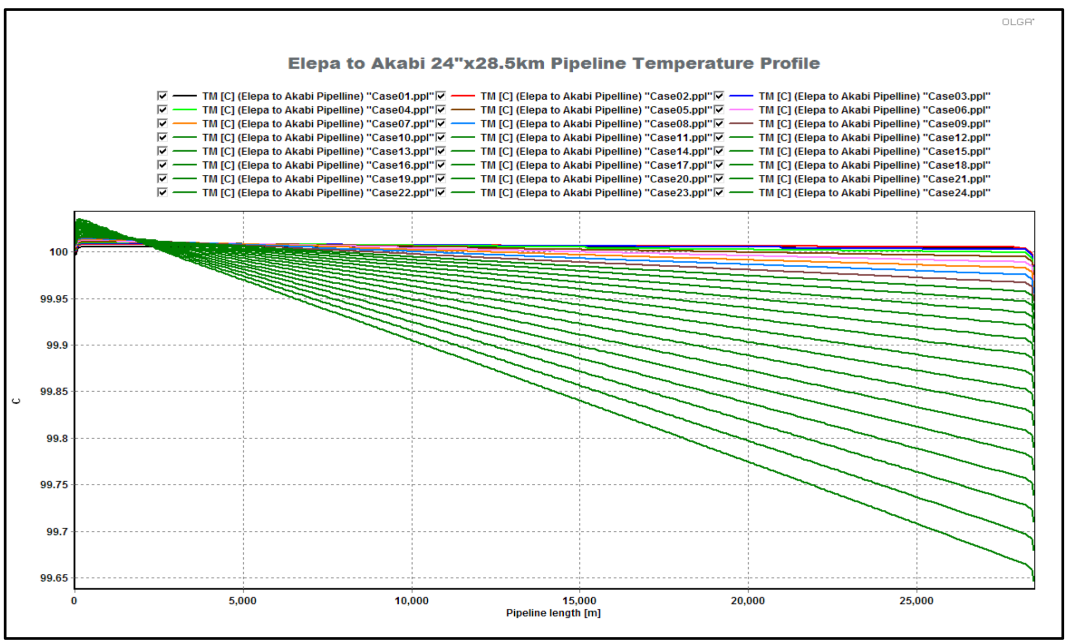

Fig 4.11: Temperature Profile Sensitivity for 24" ELEP to AKAB NAG Pipeline

Table 4.9: Case 3_ BBOU to ELEP 16" pipeline and ELEP to AKAB 24" Pipeline.

	Flowrate (MMscfd)	Inlet Pressure (bara)	Outlet Temp. (°C)	Gas Velocity (m/s)	Erosional Velocity Ratio (-)	Pipeline Water Content (m^3)	Pipeline Condensate Content (m^3)	Total Liquid Holdup (m^3)	Water Volume Flow (m^3/d)	Condensate Volume Flow (m^3/d)	Total Liquid Volume Flow (m^3/d)
BBOU to ELEP	180	82.936	56.58	8.340	0.540	0.759	51.970	52.729	16.810	1300.820	1317.630
Alepa to AKAB	300	75.901	76.82	7.750	0.490	7.963	676.290	684.253	55.840	4945.390	5001.230

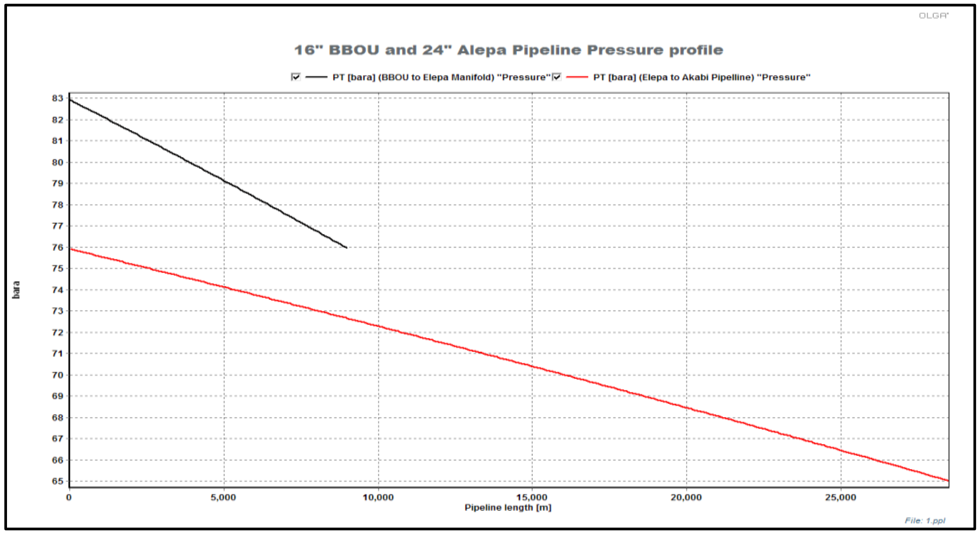

Fig 4.12: Pressure Profile Sensitivity for 16" BBOU to ELEP and 24" ELEP to AKAB NAG Pipelines

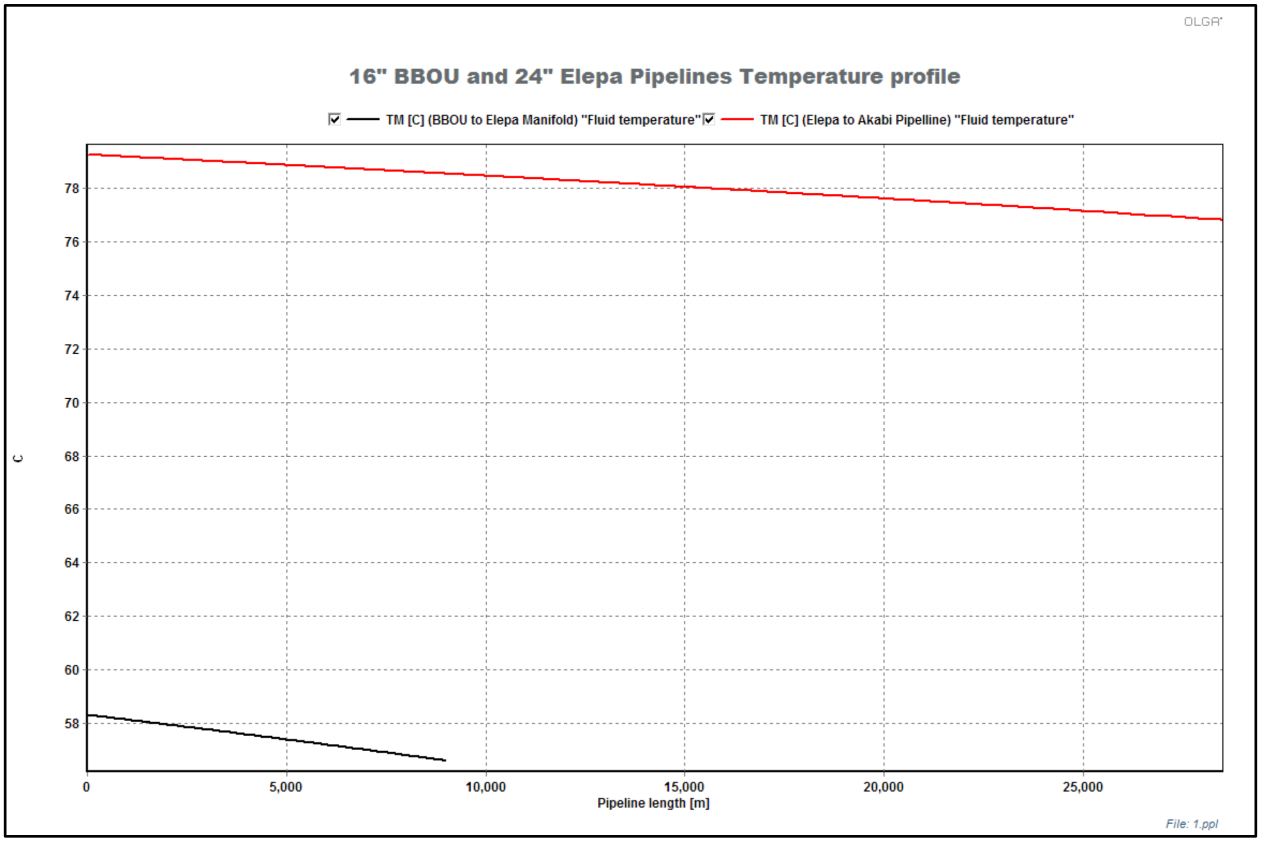

Fig 4.13: Temperature Profile Sensitivity for 16" BBOU to ELEP and 24" ELEP to AKAB NAG Pipelines

Table 4.10: Case 4_ BBOU to ELEP 16" pipeline at 300 MMscfd.

Flowrate (MMscfd)	Inlet Pressure (bara)	Outlet Temperature (°C)	Gas Velocity (m/s)	Erosional Velocity Ratio (-)	Pipeline Water Content (m^3)	Pipeline Condensate Content (m^3)	Total Liquid Holdup (m^3)	Water Volume Flow (m^3/d)	Condensate Volume Flow (m^3/d)	Total Liquid Volume Flow (m^3/d)
5	65.3010	58.2714	1.278	0.016	138.40	81.414	219.813	0.390	35.498	35.888
10	65.2892	58.2726	1.554	0.033	41.35	73.363	114.716	0.780	70.995	71.775
30	65.4609	58.2343	2.659	0.098	6.73	67.510	74.243	2.340	213.017	215.358
50	65.9682	58.1292	3.738	0.163	3.62	54.651	58.271	3.900	355.202	359.102
70	66.7608	57.9670	4.299	0.228	2.51	52.821	55.330	5.740	497.657	503.397
90	67.8190	57.7595	5.081	0.293	1.44	52.915	54.359	7.831	640.488	648.318
110	69.1640	57.5025	6.040	0.357	1.20	52.441	53.643	10.687	783.820	794.507
130	70.7803	57.2066	7.016	0.420	1.00	51.854	52.854	13.092	927.737	940.830
150	72.6763	56.8691	7.970	0.482	0.84	50.361	51.198	14.800	1072.371	1087.171
170	74.8006	56.4988	8.863	0.542	0.72	48.604	49.327	15.922	1217.765	1233.686
190	77.0855	56.0977	9.721	0.601	0.65	46.818	47.466	16.889	1363.850	1380.738
210	79.5549	55.6729	10.548	0.659	0.60	45.091	45.687	17.941	1510.747	1528.689
230	82.1974	55.2291	11.342	0.716	0.56	43.505	44.064	19.119	1658.509	1677.628
250	84.9962	54.7694	12.103	0.772	0.53	42.035	42.566	20.402	1807.082	1827.484
270	87.9472	54.2966	12.829	0.827	0.51	40.711	41.221	21.760	1956.545	1978.305
290	91.0627	53.8004	13.523	0.881	0.49	39.578	40.071	23.167	2106.864	2130.031
300	92.6956	53.5451	13.857	0.908	0.49	39.092	39.578	23.885	2182.404	2206.288

Table4.11: Case 4_ ELEP to AKAB 24" pipeline at 300 MMscfd.

Flowrate (MMscfd)	Inlet Pressure (bara)	Outlet Temperature (°C)	Gas Velocity (m/s)	Erosional Velocity Ratio (-)	Pipeline Water Content (m^3)	Pipeline Condensate Content (m^3)	Total Liquid Holdup (m^3)	Water Volume Flow (m^3/d)	Condensate Volume Flow (m^3/d)	Total Liquid Volume Flow (m^3/d)
5	65.1572	58.2302	0.368	0.007	2704.45	450.808	3155.258	0.390	35.495	35.885
10	65.1546	58.2332	0.737	0.014	996.99	587.442	1584.429	0.780	70.989	71.769
30	65.1914	58.1863	1.957	0.043	156.04	526.137	682.181	2.340	212.998	215.338
50	65.2978	58.0534	2.442	0.072	65.59	519.736	585.323	3.900	355.152	359.051
70	65.4717	57.8462	2.925	0.100	35.94	516.043	551.984	5.459	497.546	503.006
90	65.7363	57.5697	3.407	0.129	21.30	462.272	483.568	7.018	640.277	647.295
110	66.0915	57.2190	3.883	0.157	17.75	423.251	441.002	8.576	783.426	792.002
130	66.5445	56.8039	4.258	0.186	15.95	397.161	413.113	10.133	927.081	937.214
150	67.1117	56.3164	4.511	0.214	14.58	379.693	394.274	11.689	1064.318	1076.008
170	67.7549	55.7754	4.719	0.242	13.41	374.141	387.554	14.085	1216.244	1230.329
190	68.4202	55.1979	5.068	0.271	12.56	378.663	391.227	16.884	1361.785	1378.669
210	69.1501	54.5785	5.429	0.299	11.32	382.305	393.624	21.253	1507.991	1529.244
230	69.9420	53.9219	5.777	0.327	9.99	385.155	395.147	23.683	1654.897	1678.580
250	70.7935	53.2337	6.151	0.354	8.62	387.403	396.020	23.394	1802.546	1825.939
270	71.7045	52.5162	6.546	0.382	7.98	388.874	396.858	26.540	1950.964	1977.504
290	72.6700	51.7575	6.956	0.410	7.42	389.998	397.415	29.586	2100.151	2129.736
300	73.1718	51.3658	7.163	0.424	7.16	390.487	397.649	30.778	2175.068	2205.846

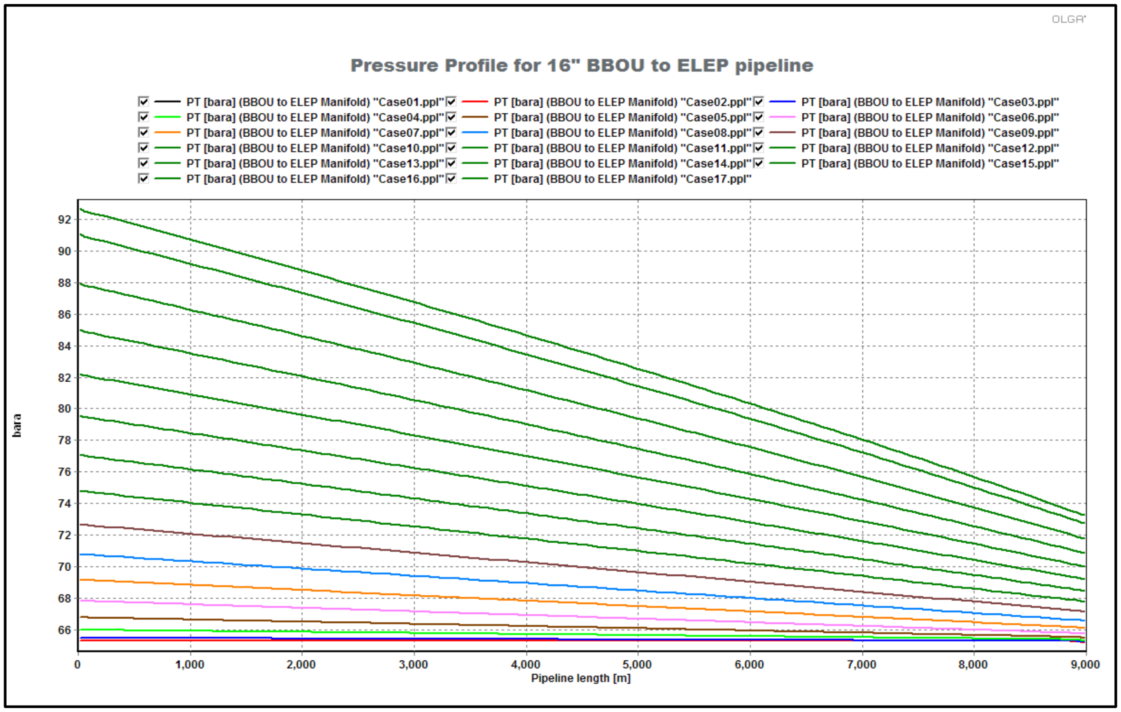

Fig 4.14: Pressure profile for 16" BBOU to ELEP pipeline (300 MMscfd)

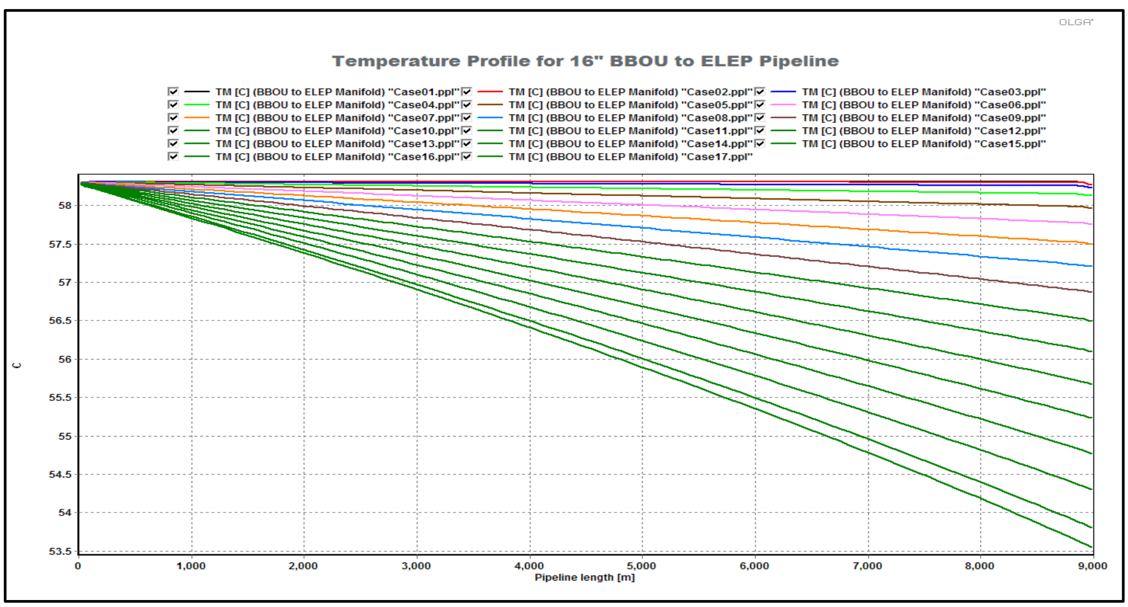

Fig 4.15: Temperature profile for 16" BBOU to ELEP pipeline (300 MMscfd)

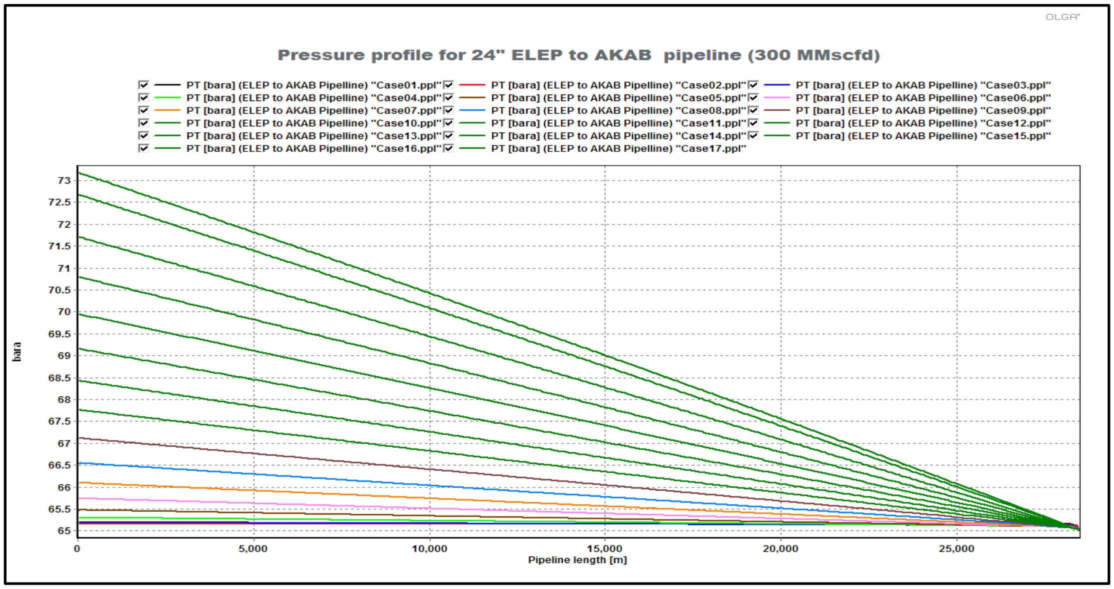

Fig 4.16: Pressure profile for 24" ELEP to AKAB pipeline (300 MMscfd)

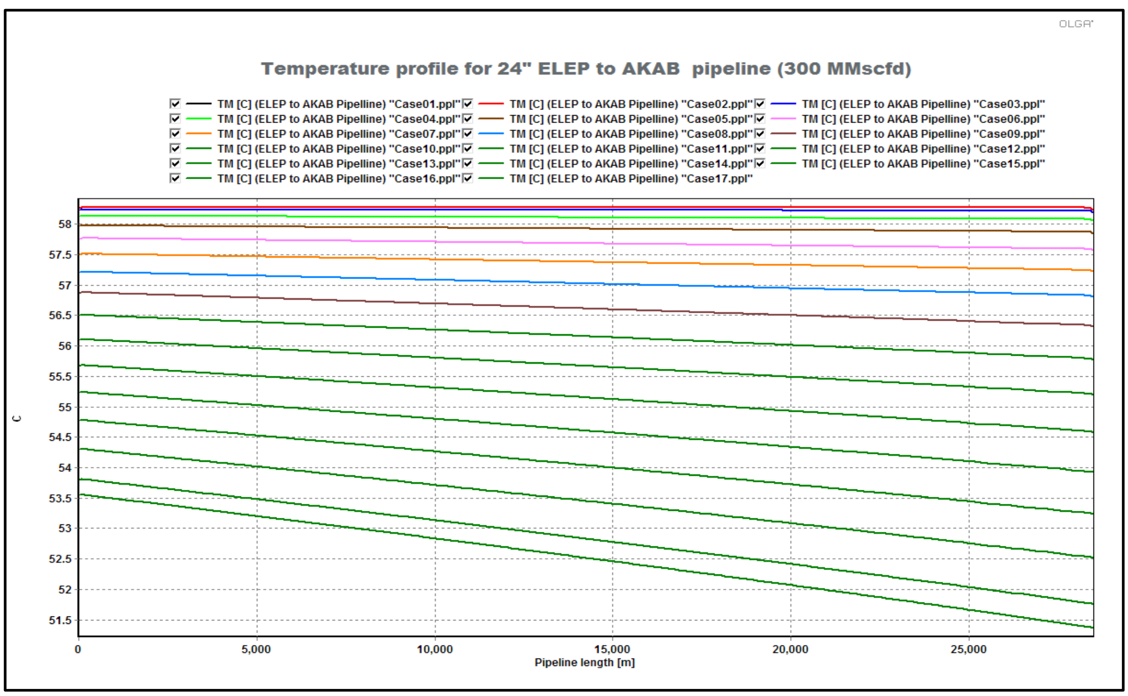

Fig 4.17: Temperature profile for 24" ELEP to AKAB pipeline (300 MMscfd)

5 Flow Regime and Slug Analysis

5.1 Case 1: Flow From BBOU Field Only

5.1.1 Flow Regime for 16" x 9 km BBOU to ELEP pipeline

The steady state flow regime for the BBOU to ELEP pipeline section is shown in the following figures for various turndown rates between 10 – 60 MMSCFD and 70 -180 MMscfd.

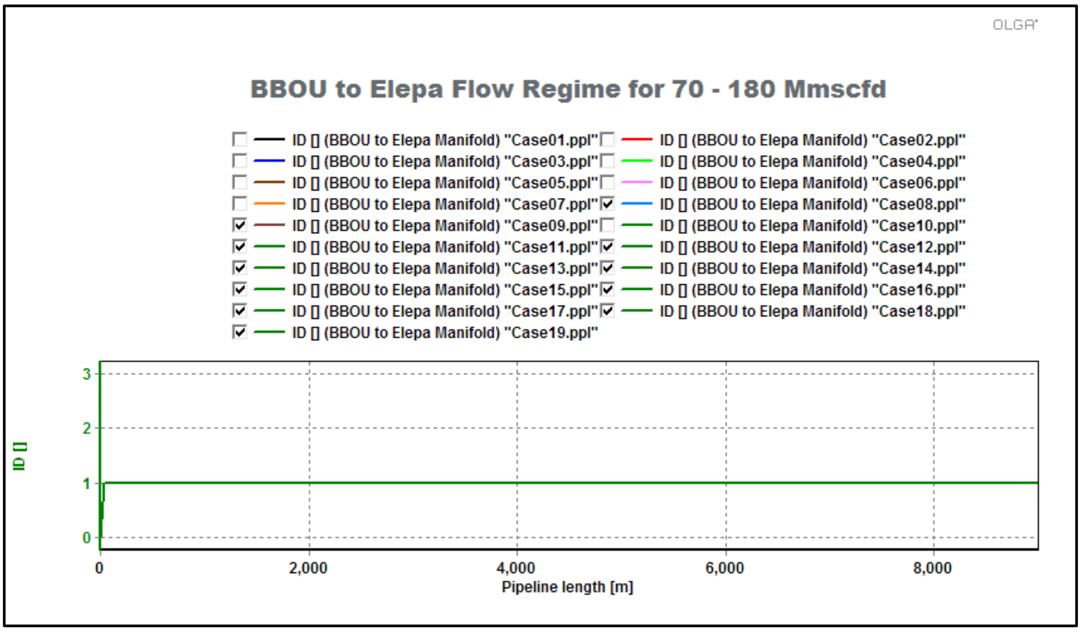

Fig 5.1: Flow Regime Indicator Plot at 70 – 180 MMscfd for BBOU to ELEP NAG pipeline

From the flow regime indicator, it is obvious that all part of the BBOU to ELEP pipeline is in stratified flow (ID=1).

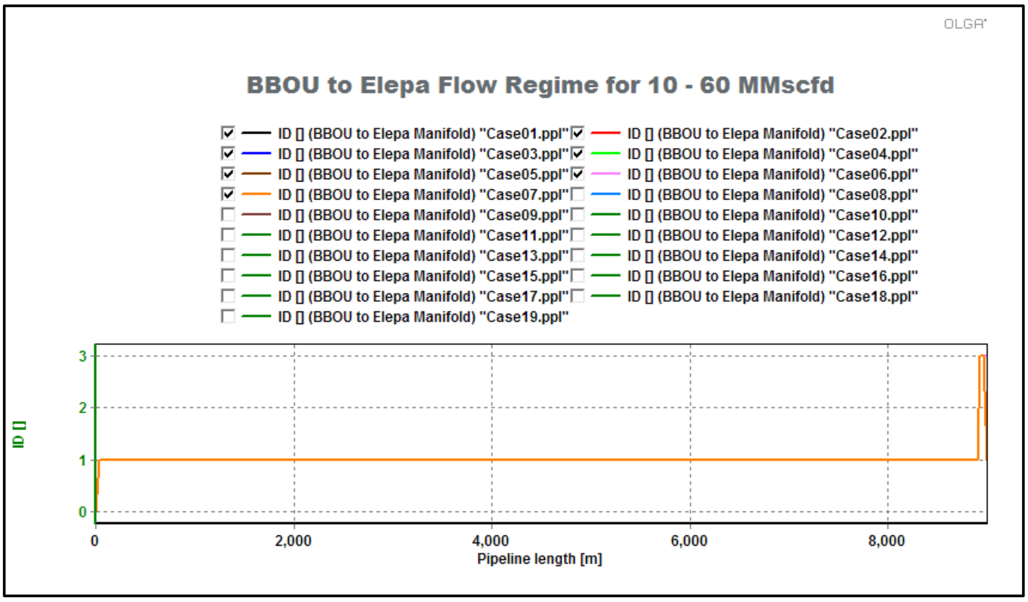

Fig 5.2: Flow Regime Indicator Plot at 10 – 60 MMscfd for BBOU to ELEP NAG pipeline

From the flow regime indicator, it is obvious that the end part of the BBOU to Elepa pipeline is in hydrodynamic slug flow (ID=3). Slugging tendencies increase as flowrate decreases.

5.1.2 Flow Regime for 24" x 28.5 km ELEP to AKAB pipeline

The steady state flow regime for the ELEP to AKAB pipeline section is shown in the following figures for various turndown rates between 10 – 180 MMscfd.

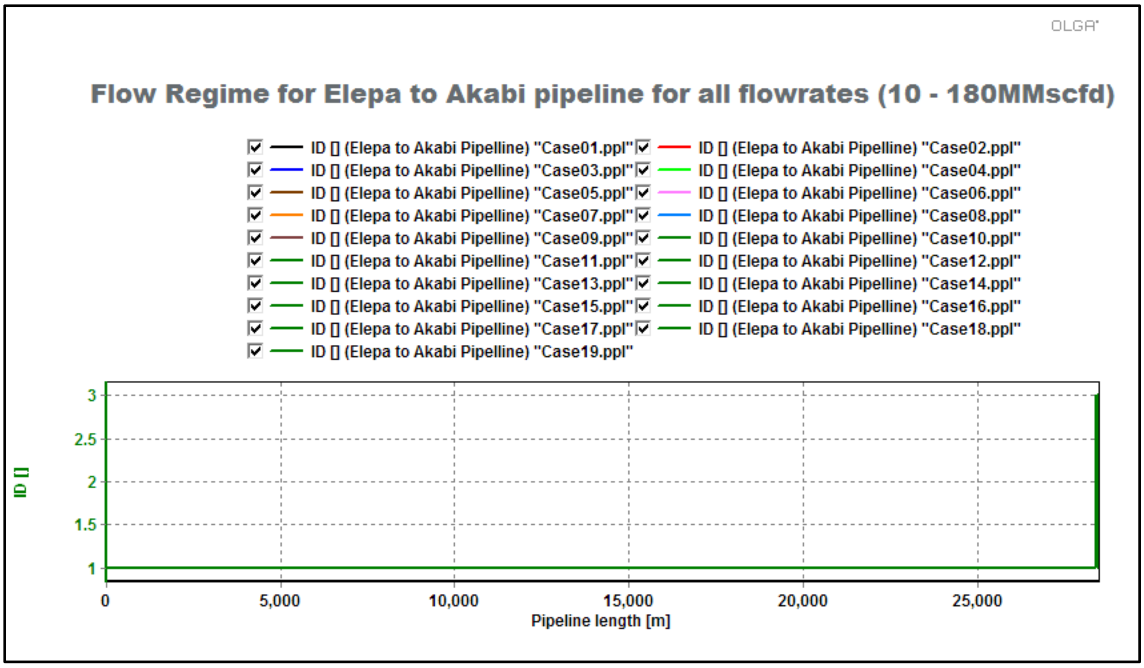

Fig 5.3: Flow Regime Indicator Plot at 10 – 180 MMscfd for ELEP to AKAB NAG pipeline

From the flow regime indicator, it is obvious that the end part of the Elepa to Akabi pipeline is in hydrodynamic slug flow (ID=3). Slugging tendencies increase as flowrate decreases.

5.2 Case 2: 120 MMscfd flow from ELEP field

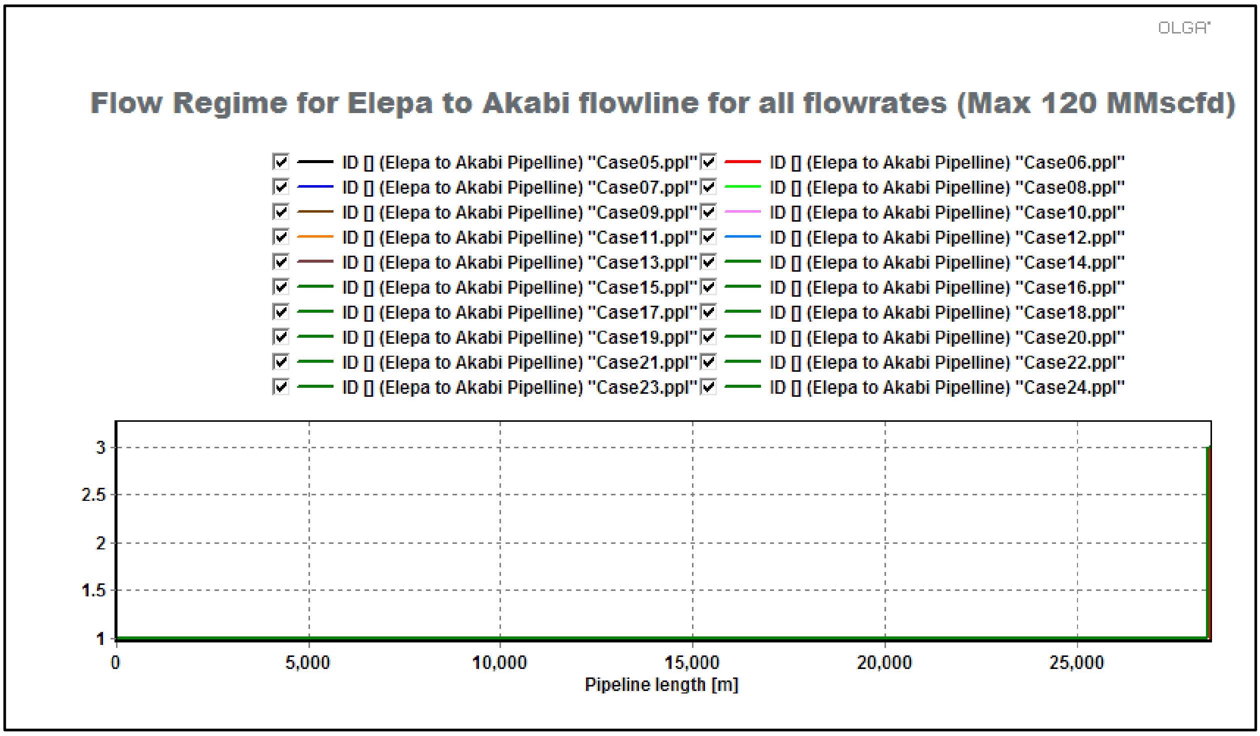

Fig 5.4: Flow Regime Indicator Plot at 10 – 180 MMscfd for ELEP to AKAB NAG Pipeline

From the flow regime indicator, it is obvious that the end part of the Elepa to Akabi pipeline is in hydrodynamic slug flow (ID=3). Slugging tendencies increase as flowrate decreases.

5.3 Case 3: Combined Flow from BBOU(180 MMscfdd) and ELEP (120 MMscfd)

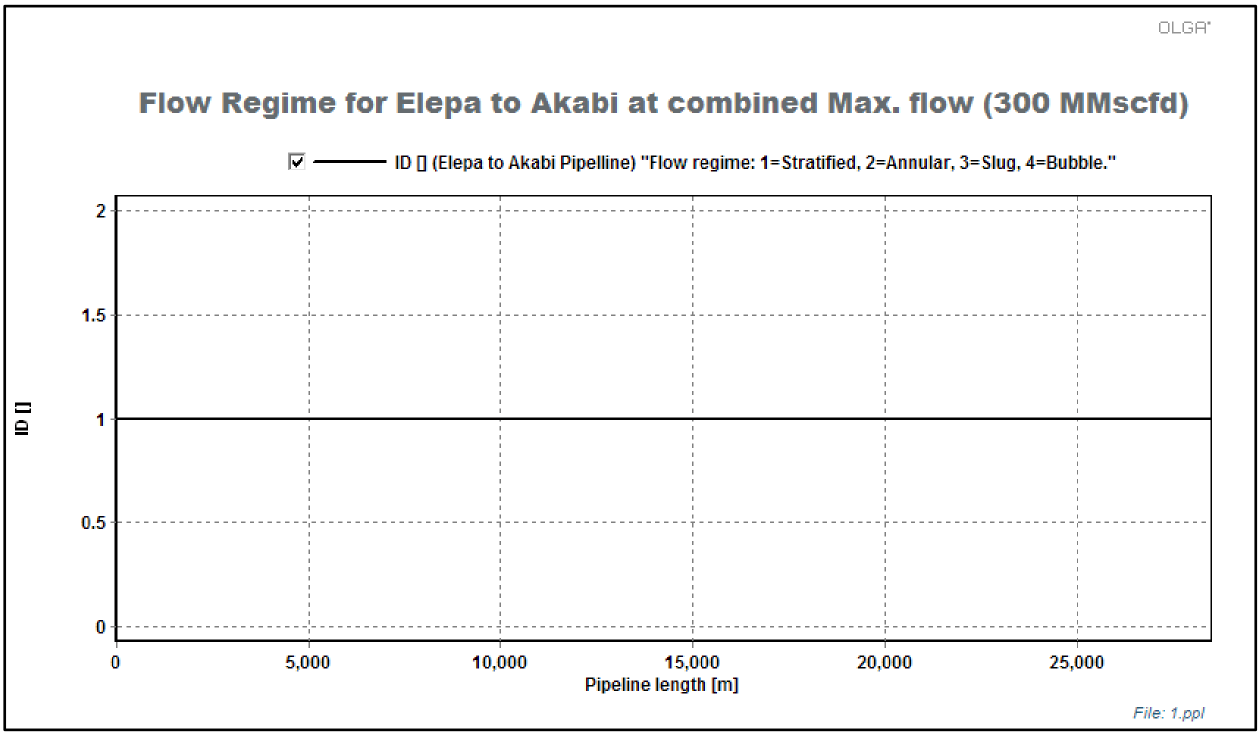

Fig 5.5: Flow Regime Indicator Plot at 300 MMscfd for ELEP to AKAB NAG pipeline

From the flow regime indicator, it is obvious that all part of the BBOU to ELEP pipeline is in stratified flow (ID=1).

6 Transient Simulations

6.1 Shutdown Simulation

Pipeline shutdown was performed in order to check the possibility of hydrate formation during shutdown period.

Pipelines were operated for 3 hours then shutdown and allowed to cool down for a period of one day (1 day). After the shut-in period it was observed that the fluid temperature for the whole section of the pipeline is above the hydrate formation temperature as shown in figures 6.1 and 6.2.

Hence, there is no risk of hydrate formation during this shutdown period.

6.1.1 Shutdown for BBOU at 300MMscfd

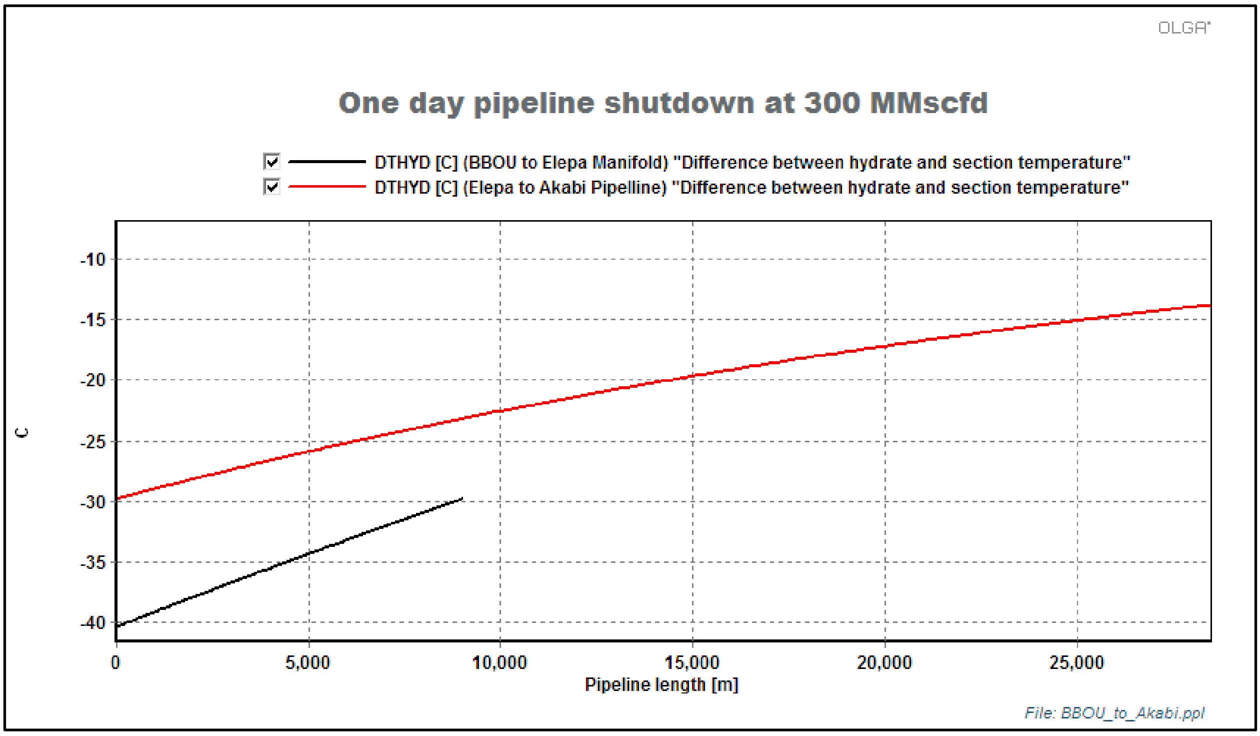

Fig 6.1: Difference between hydrate and section temperature for BBOU to ELEP and ELEP to AKAB NAG pipelines.

6.1.2 Shutdown for combined BBOU and ELEP

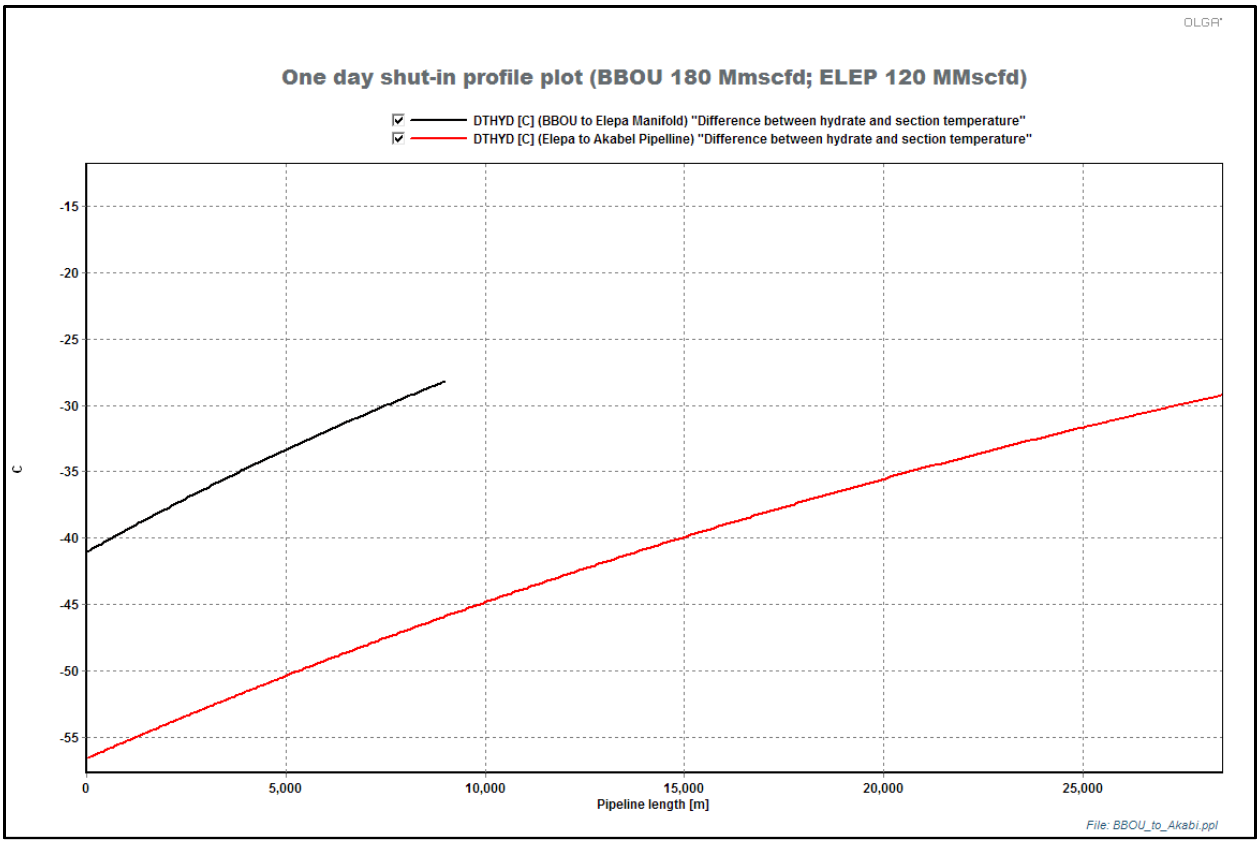

Fig 6.2: Profile plot for difference between hydrate and section temperature for BBOU to ELEP and ELEP to AKAB NAG pipelines.

6.2 Ramp-Up Simulation

From the minimum turndown result, ramp-up was done over 3 and 6 hours in an effort to bring the pipeline back to the peak flowrate and determine the liquid rate and surge volume into Akabel processing facility.

Four ramp studies were simulated

- Ramp up of BBOU pipeline from minimum turndown of 10 MMscfd to peak production of 180 MMscfd in 3 hours
- Ramp up of BBOU pipeline from minimum turndown of 10 MMscfd to peak production of 180 MMscfd in 6 hours
- Ramp up of BBOU pipeline from minimum turndown of 10 MMscfd to peak production of 300 MMscfd in 3 hours
- Ramp up of BBOU pipeline from minimum turndown of 10 MMscfd to peak production of 300 MMscfd in 6 hours

6.2.1 Ramp up of BBOU stream from 10 MMscfd to 180 MMscfd in 3 hrs

Fig 6.3 shows the liquid arrival rate at Akabel processing plant for a ramp-up from 10 MMscfd to 180 MMscfd within 3 hours. The maximum liquid ramp-up is 1508.22 m^3/day.

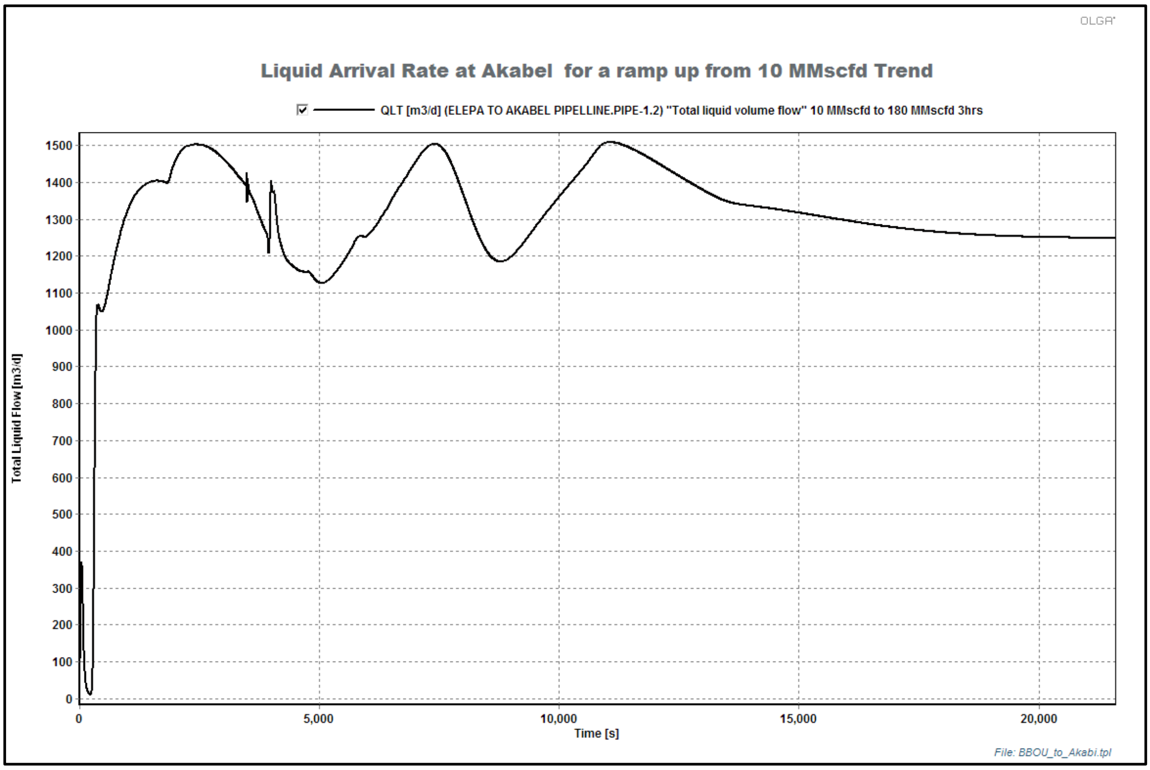

Fig 6.3: Liquid rate at Akabel Processing Plant during ramp-up of BBOU stream from 10 MMscfd Turndown to 180 MMscfd in 3hrs

6.2.2 Ramp up of BBOU stream from 10 MMscfd to 180 MMscfd in 6 hrs

Fig 6.4 shows the liquid arrival rate at Akabel processing plant for a ramp-up from 10 MMscfd to 180 MMscfd within 6 hours. The maximum liquid ramp-up is 1371.38 m^3/day.

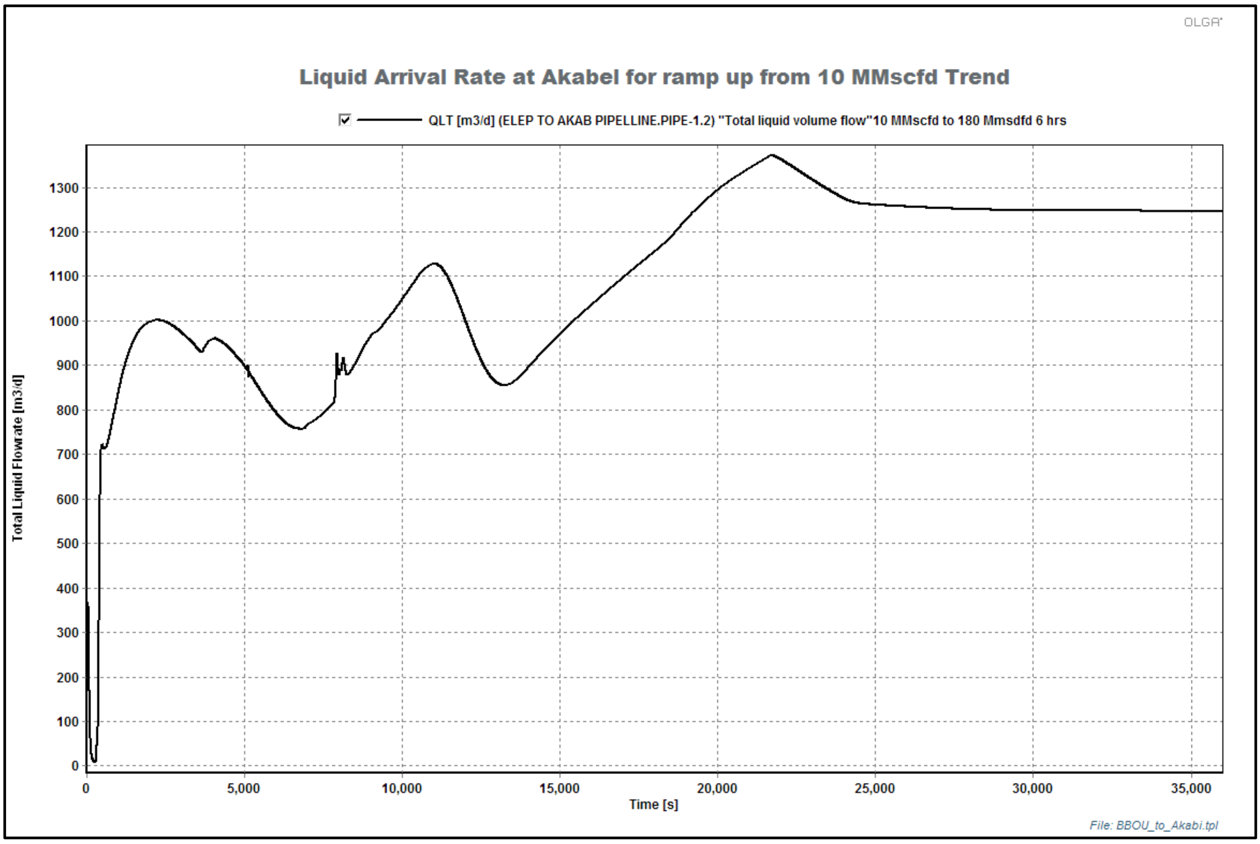

Fig 6.4: Liquid rate at Akabel Processing Plant during ramp-up of BBOU stream from 10 MMscfd Turndown to 180 MMscfd in 6hrs

The steady state arrival rate at the processing facility at full gas capacity of 180 MMscfd is 1306.811 m^3/day.

6.2.3 Ramp up of BBOU stream from 10MMscfd to 300MMscfd in 3 hrs

Fig 6.5 shows the liquid arrival rate at Akabel processing plant for a ramp-up from 10 MMscfd to 300 MMscfd within 3 hours. The maximum liquid ramp-up is 2389.92m^3/day.

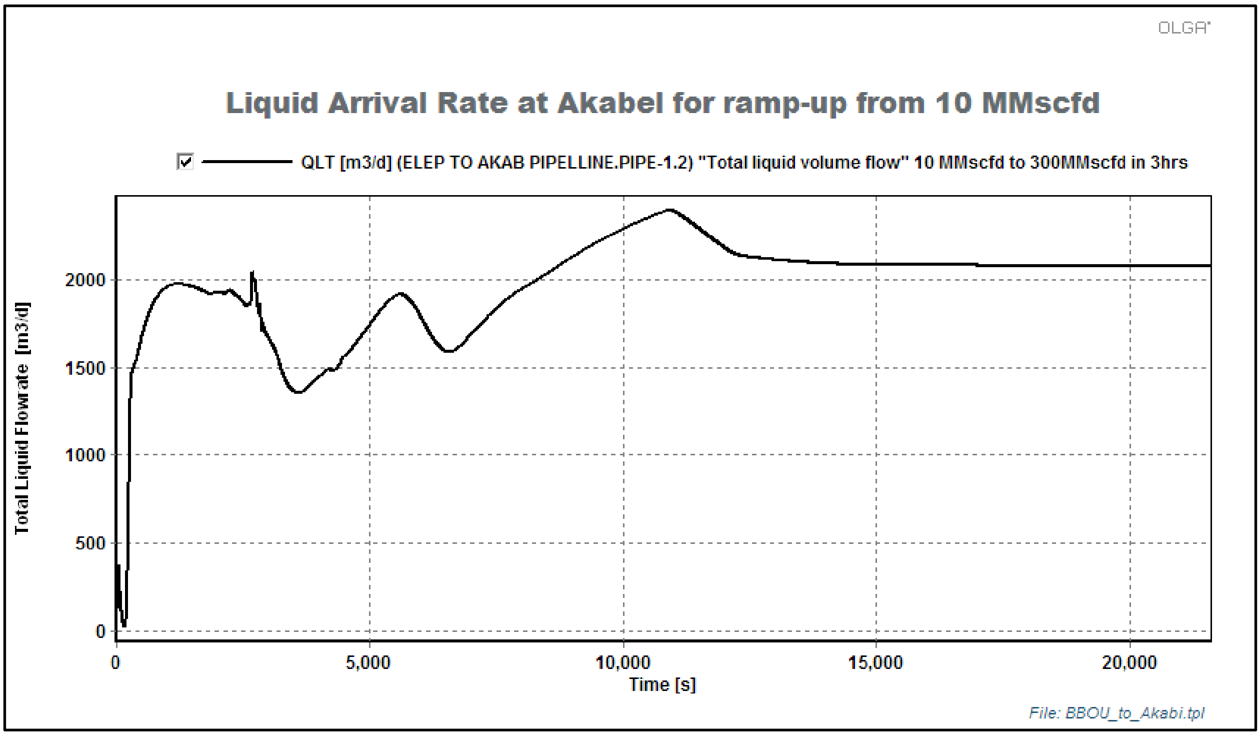

Fig 6.5: Totl Liquid rate at Akabel Processing Plant during ramp-up of BBOU stream from 10 MMscfd Turndown to 300 MMscfd in 3hrs

6.2.4 Ramp up of BBOU stream from 10MMscfd to 300MMscfd in 6 hrs

Fig 6.6 shows the liquid arrival rate at Akabel processing plant for a ramp-up from 10 MMscfd to 300 MMscfd within 6 hours. The maximum liquid ramp-up is 2156.96m^3/day.

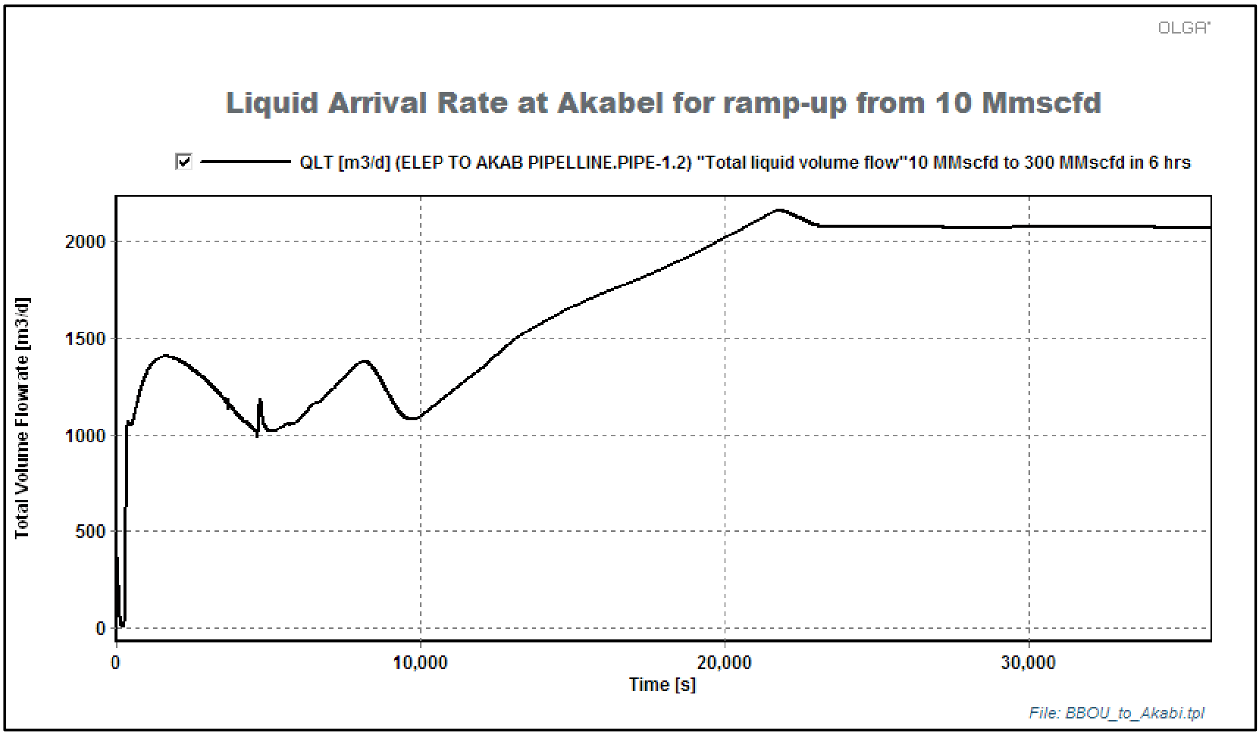

Fig 6.6: Total Liquid rate at Akabel Processing Plant during ramp-up of BBOU stream from 10 MMscfd Turndown to 300 MMscfd in 6hrs

The steady state arrival rate at the processing facility at full gas capacity of 300 MMscfd is 2206.288m^3/day

6.3 Start-up Simulations

Start-up simulation for all pipelines was carried out in order to determine the maximum liquid volume out of the pipelines and the liquid surge volumes resulting from the start-up following a period of shutdown.

6.3.1 BBOU Re-Start (BBOU: 300 MMscfd)

BBOU to AKAB NAG pipeline was re-started using the re-start file from one-day shut-in simulation. After one-day shut-in, the pipeline was restarted and operated for another one hour.

The maximum total liquid volume flow out of the pipeline during start-up is 3250.26m^3/day.

The total liquid volume flow into CPF during Benisede start-up is shown in figure 6.7 below:

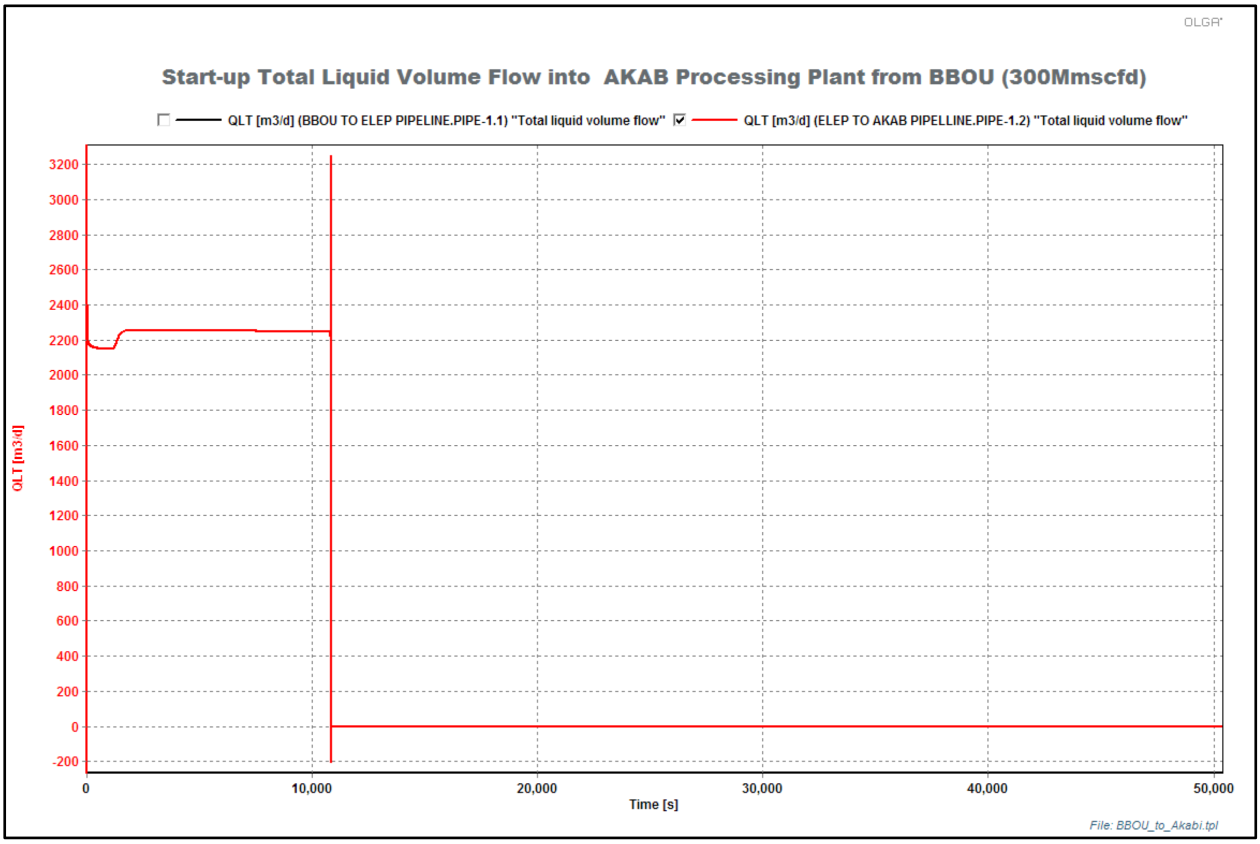

Fig 6.7: Total Liquid rate at Akabel Processing Plant during Restart of BBOU Pipeline after shutdown.

6.3.2 Restart of Combined Flow From BBOU and AKAB (BBOU 180MMscfd; AKAB 120MMscfd)

BBOU to AKAB NAG pipeline was re-started using the re-start file from one-day shut-in simulation. After one-day shut-in, the pipeline was restarted and operated for another one hour.

The maximum total liquid volume flow out of the pipeline during start-up is 5105.98m^3/day.

The total liquid volume flow into CPF during Benisede start-up is shown in figure 6.8 below:

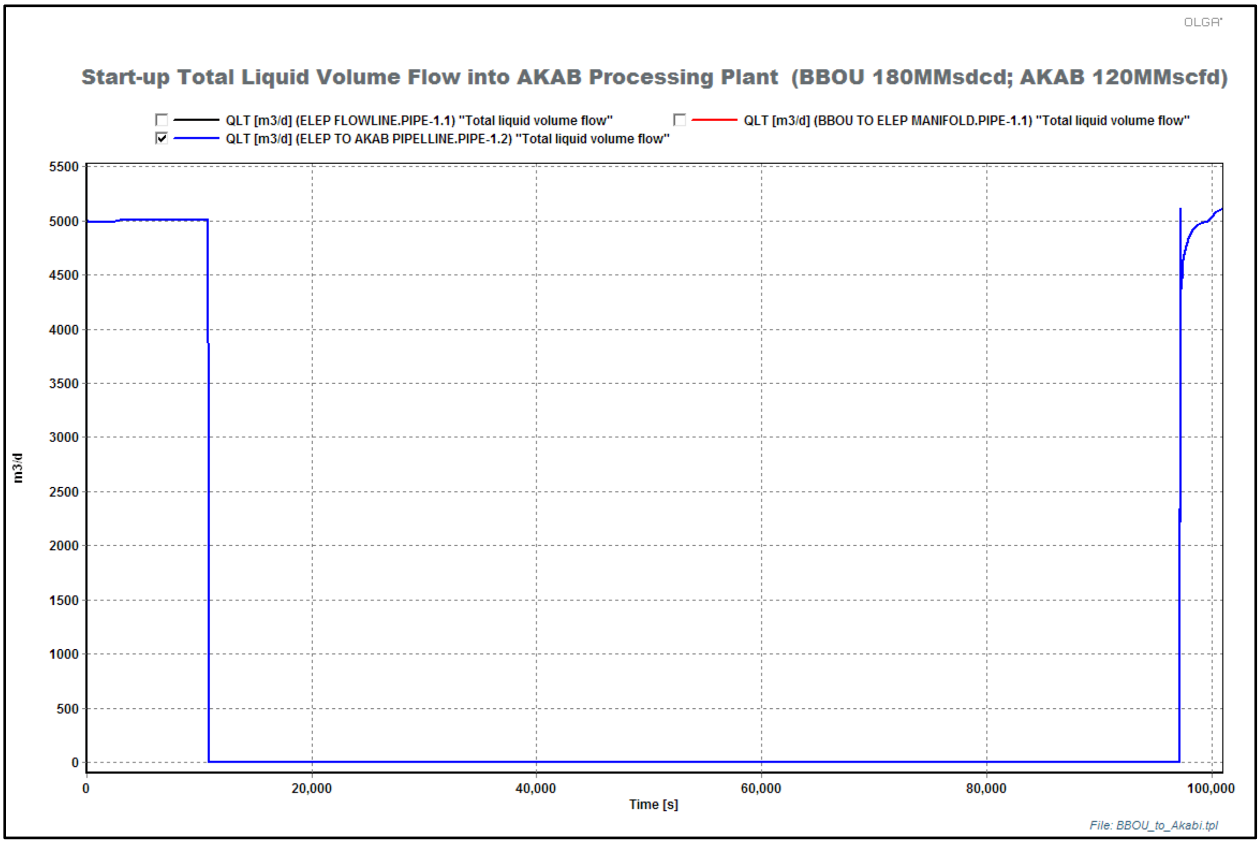

Fig 6.8: Total Liquid rate at Akabel Processing Plant during Restart of BBOU Pipeline and ELEP to AKAB Pipeline after shutdown.

6.4 Pigging operation

Pigging simulation of the pipelines were done to determine the following:

1. Optimum range of gas flowrate that will meet the velocity range specified by the pig tool manufacturer
2. Duration of the actual pigging operation as a function of gas flowrate

Different pigging gas rates were studied.

The following table shows the pig data used

Table 6.1: Pig Details for OLGA Simulation

Property	Olga Default Value (Efficient)	Source
Static force (N)	1000	Default
Wall friction (Ns/m)	1000	Default
Linear friction (Ns/m)	10	Default
Quadratic friction (Ns/m2)	0	Default
Mass (kg)	350	Rosen Tools Guide
Leakage factor (-)	0	Default
Pig Velocity Range (m/s)	0.5 – 5.0	Rosen Tools Guide

APPENDIX

Appendix-1: Inlet Stream Compositions

Appendix 1	Inlet Stream Compositions	PDF Inlet Stream Composition .pdf

Key References

1. Wellhead Gas Gathering Schematics & Flow Parameters to Gas Processing Plant.

2. Field's & Water Saturated Composition

3. Process Simulation Study - Hysys

Reference-1: Wellhead Gas Gathering Schematics & Flow Parameters............

	Document Ref	Document	Information
Ref 1	EA/SPDC-BFPC UGS-GEP/BDM-PFS-001-A01 Wellhead Gas Gathering Schematics & Flow Parameters to Gas Processing Plant	EA SPDC-BFPC UGS-GEP BDM-PFS-0(	Overall Process Route- Path & Flow Parameters

Reference-2: Prod. Forecast; Field's and Water Saturated Gas Composition

	Document Ref	Document	Information
Ref 2	Production Forecast; Field's and Water Saturated Gas Composition	Prod.Frcast-Resvr & Saturated-Comps	Production Forecast; Field's Compositions/Properties; and Water Saturated Composition

Reference-3: Process Simulation Study - Hysys

	Document Ref	Document	Information
Ref 3	EA/SPDC-BFPCUGS-GEP/PRC-RPT-001-A01 Process Simulation Study - Hysys	Hysys Simulation Study.docx	Process